

Marine Navigation



Marine Navigation Pt. 1

- Introduction to Navigation
- Coordinate Systems
- Nautical Charts
- Aids to Navigation
- Tides

Marine Navigation Pt. 2

- The Magnetic Compass
- Obtaining a Fix
- Dead Reckoning
- Navigation Rules
- Electronic Instruments

What Is Navigation?

Navigation: Determination of one's position, and velocity. (Where are you?)

Guidance: Determination of the path to a destination. (Where are you going? How will you get there?)

Collision Avoidance: Getting there safely.

Types of Marine Navigation

- Dead Reckoning

Starting from a known location, track your position based on the speed and direction your vessel travels over measured periods of time.

- Piloting

Navigating by sight in restricted waters using landmarks such as geographic features and other aids to navigation with reference to a nautical chart.

- Celestial Navigation

Determining your position based on the positions of the sun, moon, stars, and other celestial objects, usually measured with a sextant.

Types of Marine Navigation

- Radio Navigation

Determining your position based on radio signals (e.g. RDF, LORAN).

- Radar Navigation

Using radar to determine direction and distance to nearby objects.

- Inertial Navigation

Use of accelerometers and gyroscopes to keep an electronic dead reckoning. Not used much on sailboats.

- Satellite Navigation

Use of satellite based electronic systems; such as GPS.

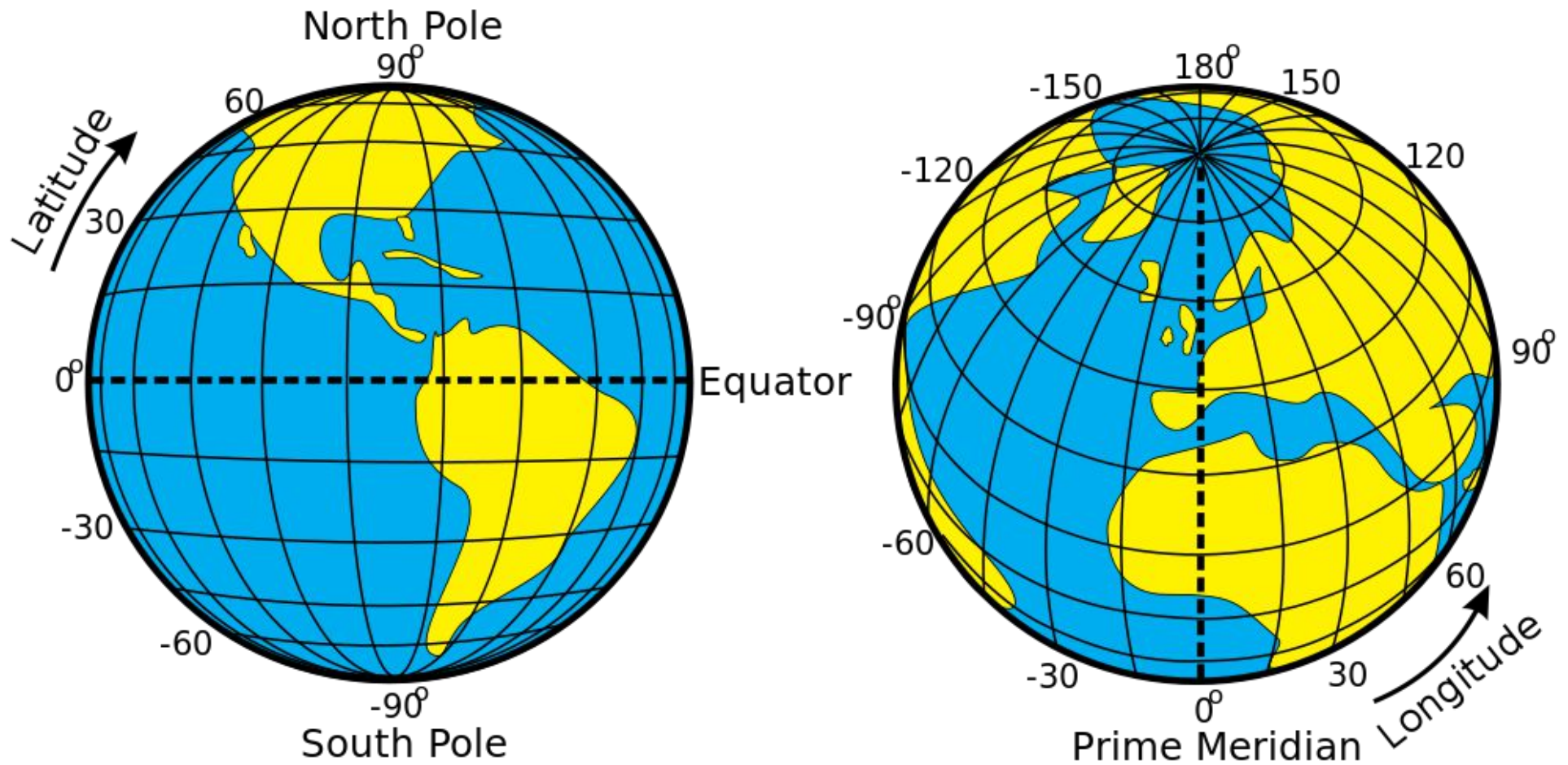
Marine Navigation

The Navigation Rules... expect prudent mariners to avail themselves of **all available means appropriate...** as to make **full appraisal of the situation.**

WARNING

The prudent mariner will not rely solely on any single aid to navigation, particularly on floating aids.

Latitude and Longitude



Parallels of **Latitude**: Position **North** or **South** of Equator

Meridians of **Longitude**: Position **East** or **West** of Prime Meridian
(Greenwich)

Specifying Latitude and Longitude

360 degrees ($^{\circ}$) in a circle
60 minutes ($'$) in 1 degree
60 seconds ($''$) in 1 minute

North & East: positive
South & West: negative

Coordinates of MIT Sailing Pavilion:

42° 21' 30.4" N, 71° 5' 15.6" W
42° 21.507' N, 71° 5.260' W
42.35845° N, 71.08776° W
42.35845, -71.08776

Latitude should be written first



Specifying Latitude and Longitude

360 degrees ($^{\circ}$) in a circle
60 minutes ($'$) in 1 degree
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Coordinates of MIT Sailing Pavilion:

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42.35845° N, 71.08776° W

42.35845, -71.08776

When specifying coordinates for use in navigation, use degrees and minutes. Be sure to include at least one decimal place for the minutes.



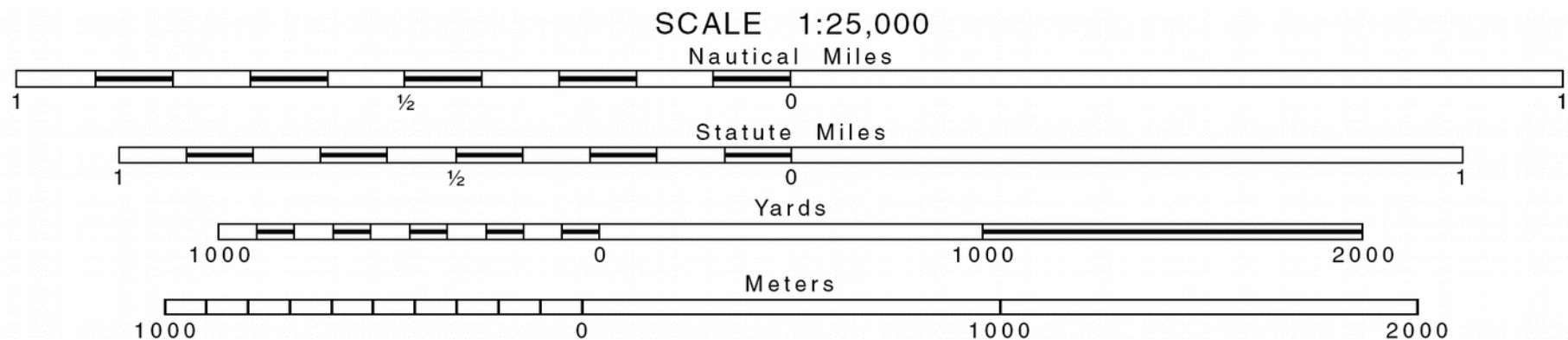
Nautical Mile

1 nautical mile = 1852 meters (exactly)

1 nautical mile $\approx$ one minute of latitude

1 nautical mile $\approx$ 1.15 statute miles

1 nautical mile $\approx$ 6076 feet

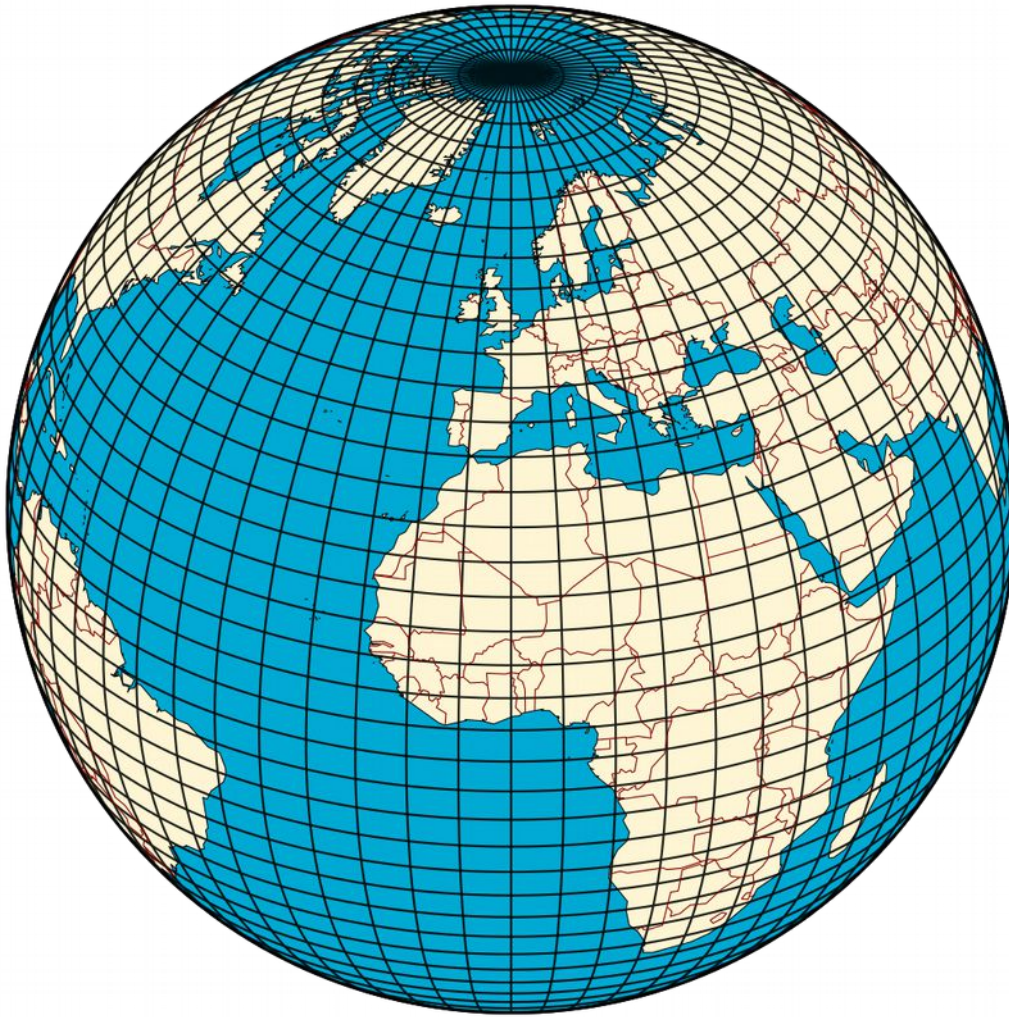


1 knot = 1 nautical mile per hour

1 knot = 1.852 kph (exactly)

1 knot $\approx$ 1.15 mph

Latitude and Longitude Distances



Latitude:

Parallels are evenly spaced.

1 minute $\approx$ 1 nm.

Longitude:

Meridians converge at poles.

1 minute $\approx \cos(\text{lat}) \times 1$ nm.

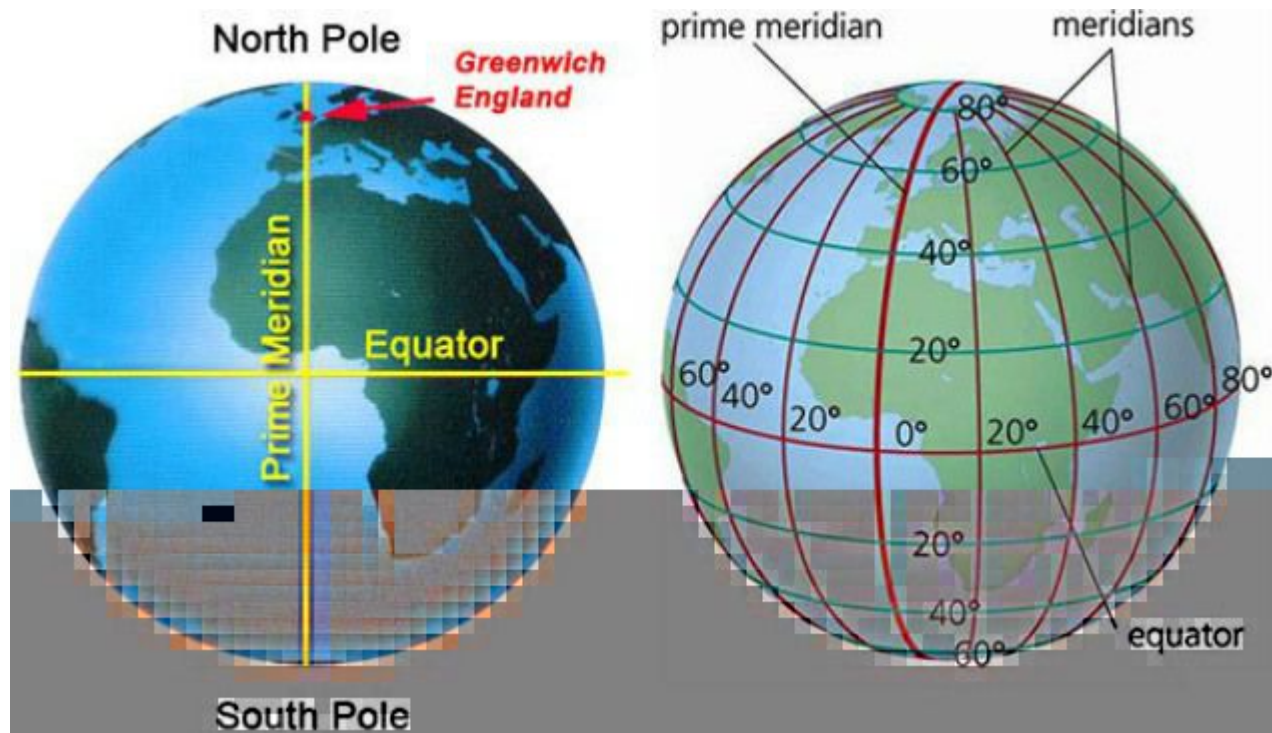
In Boston Harbor:

1 minute longitude $\approx$.74 nm.

Horizontal Datum

Coordinate system and set of reference points for assigning geographic coordinates (latitude and longitude) to physical locations on the Earth.

Location of Prime Meridian (where longitude = 0)

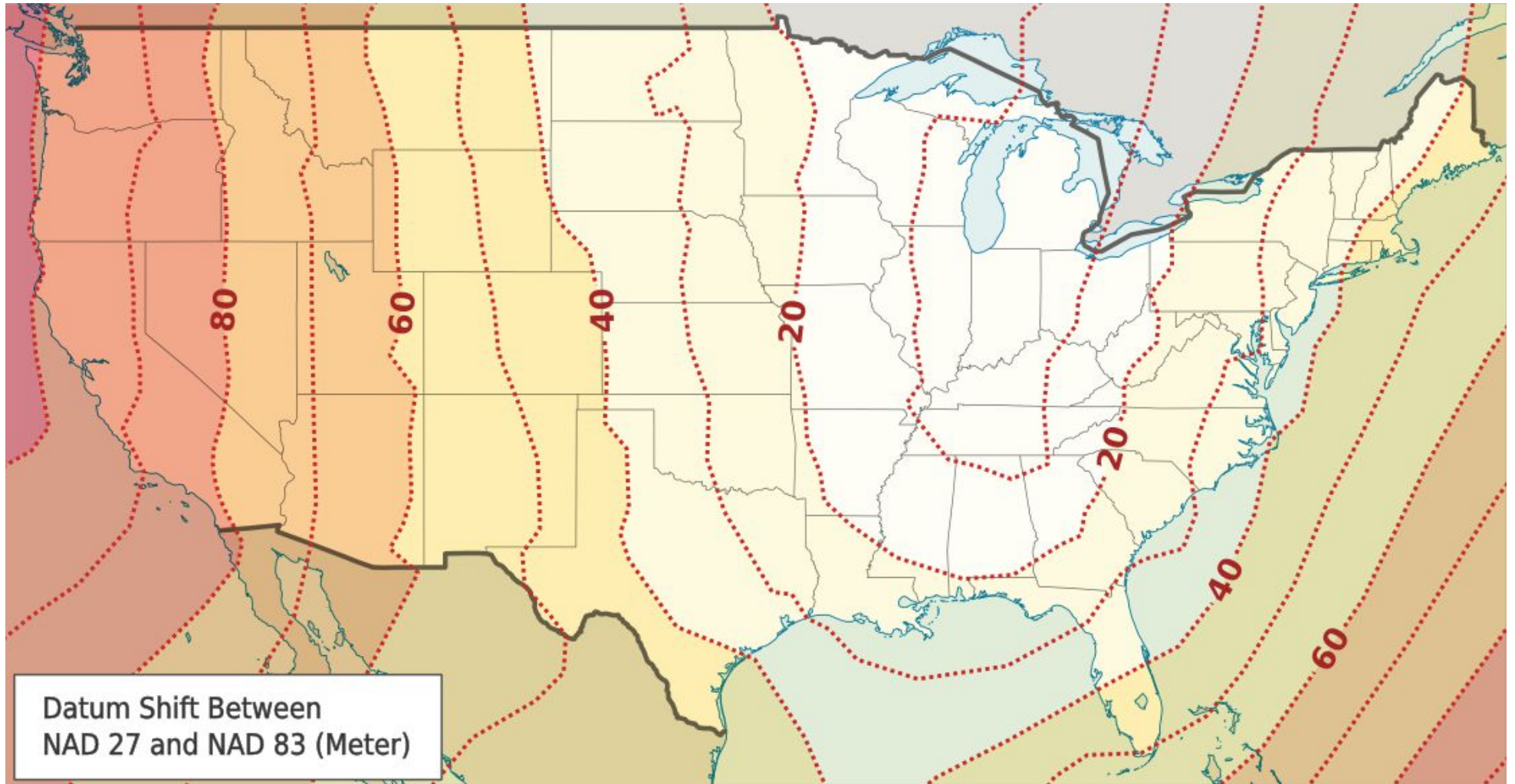


Horizontal Datum

Make sure your GPS and other navigation instruments are using the same Horizontal Datum as your charts.

- World Geodetic System 1984 (WGS84) global standard
- North American Datum 1983 (NAD83) official datum used on all U.S. Charts – varies less than 2 m from WGS84.
- North American Datum 1927 (NAD27) outdated – can vary up to 100 m from WGS84
- Hundreds of other local datums in use around the world.

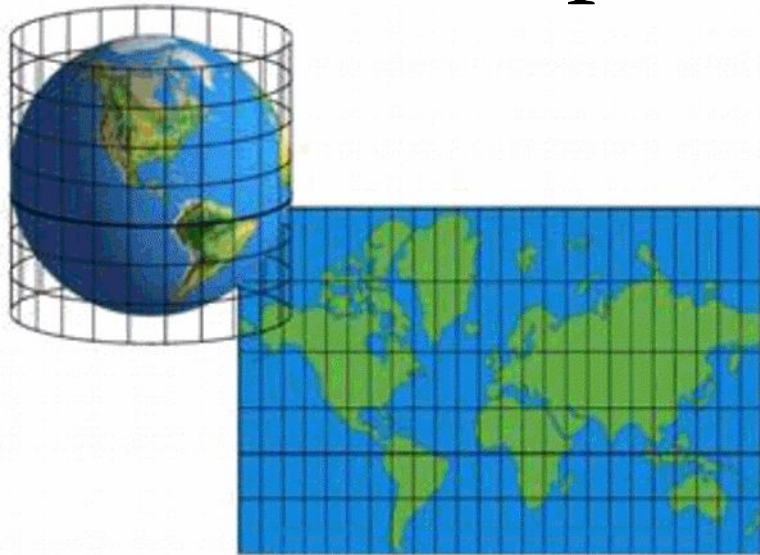
Horizontal Datum



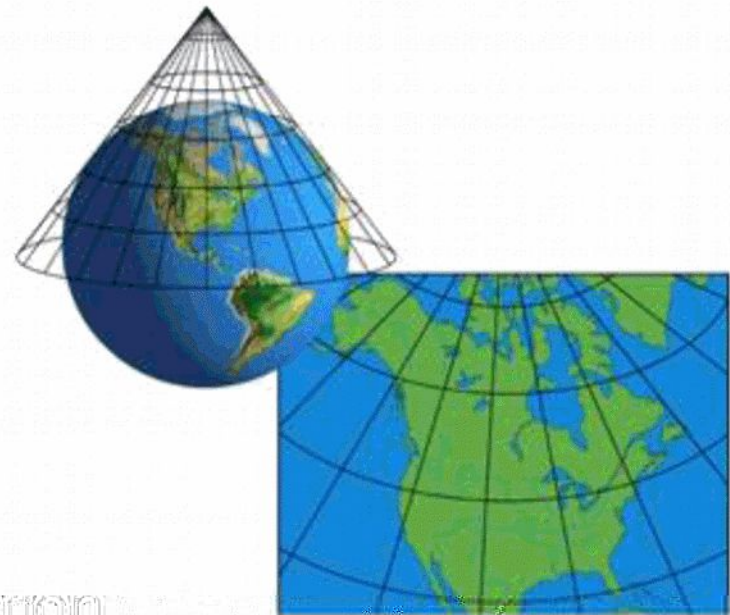
Nautical Charts

- Map Projections (Mercator, etc.)
- Chart Features (legend, scale, etc.)
- Chart Symbols
- Aids to Navigation

Map Projections



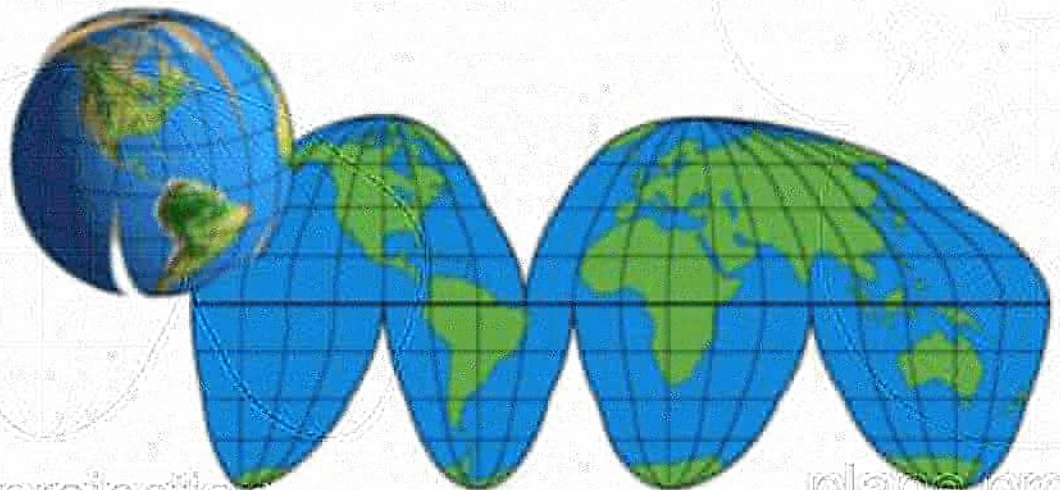
cylindrical projection



conic projection

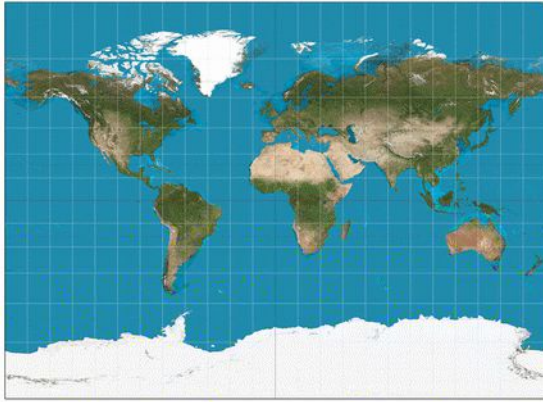


plane projection

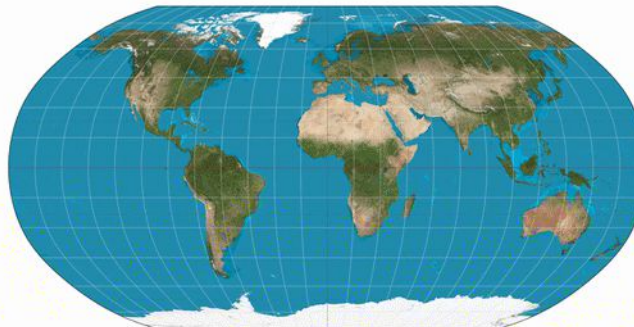


interrupted projection

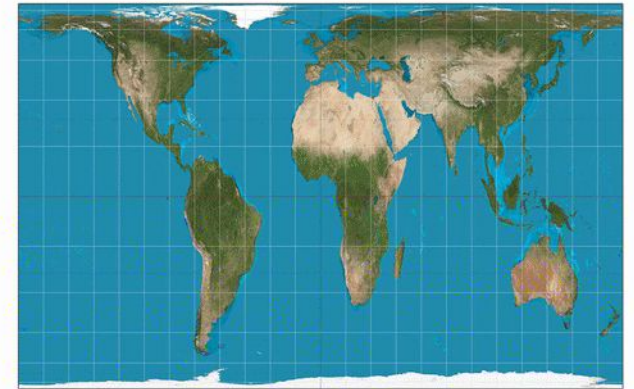
Map Projections



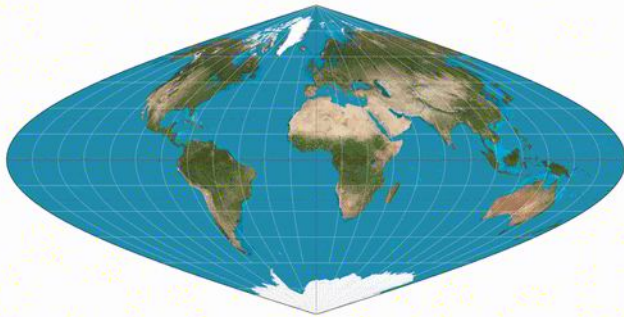
Miller Cylindrical Projection



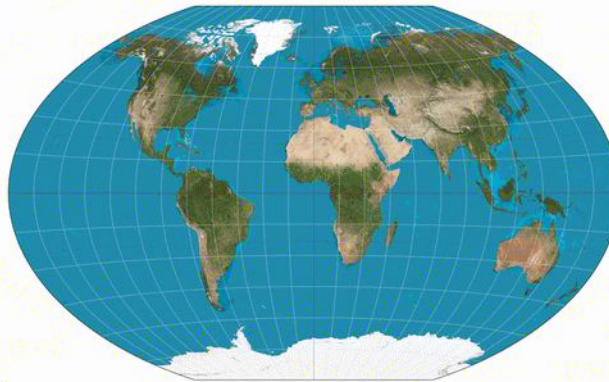
Robinson Projection



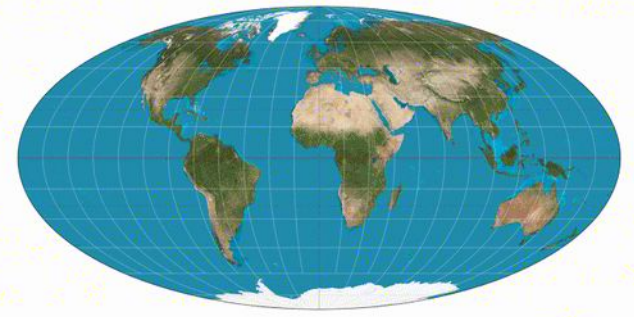
Gall-Peters Projection



Sinusoidal Equal-Area Projection

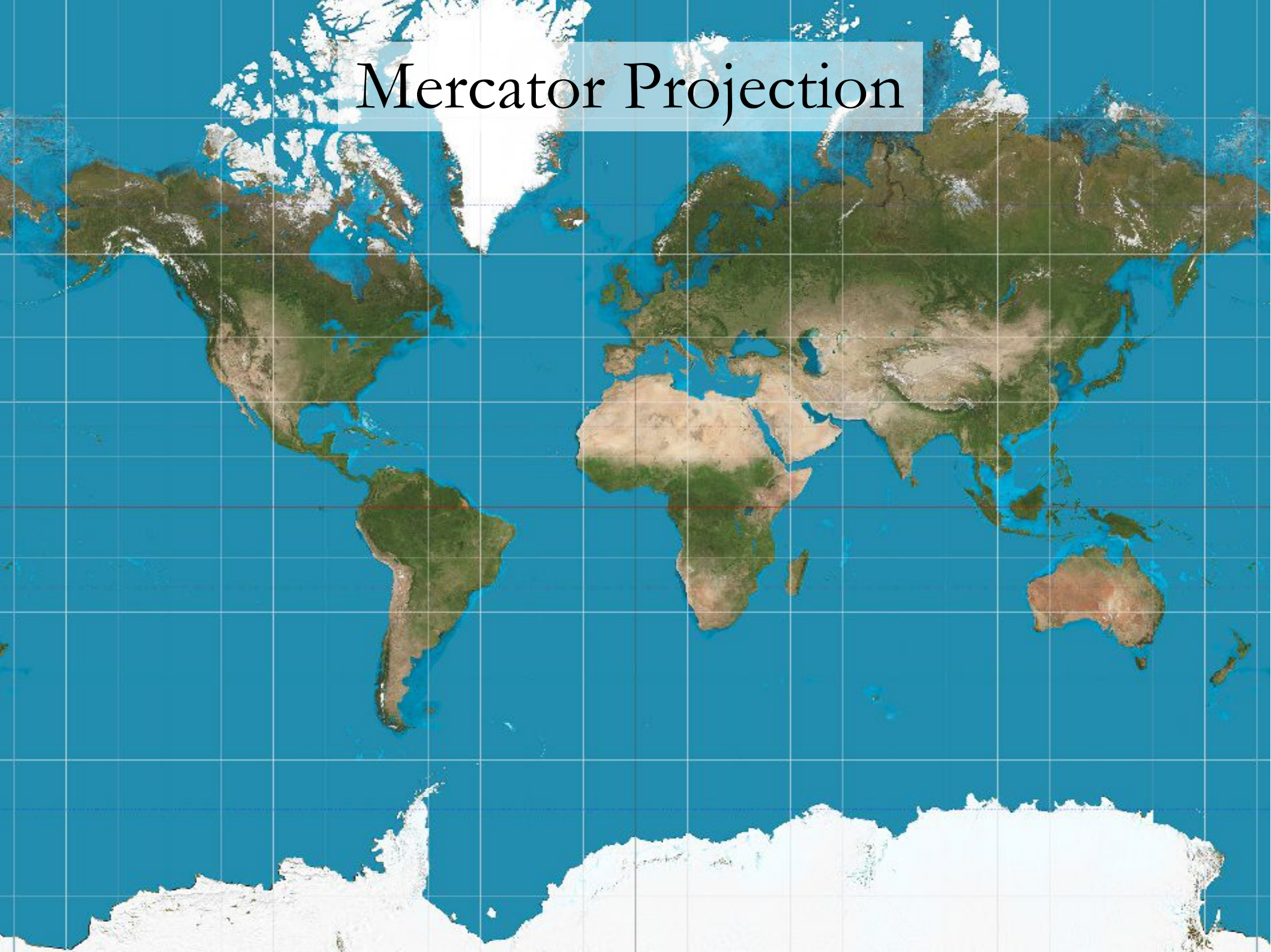


Winkel Tripel Projection



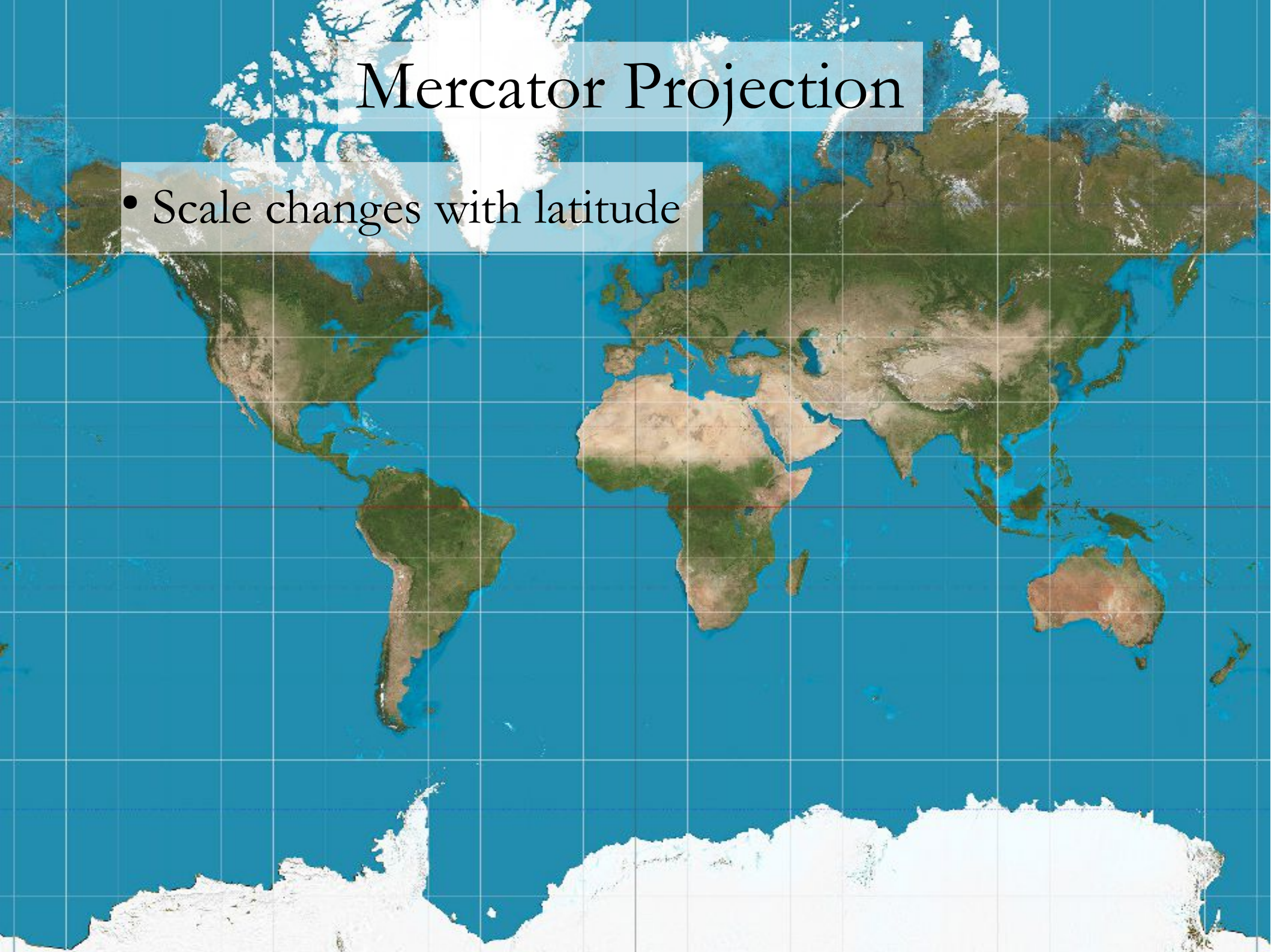
Mollweide Projection

Mercator Projection



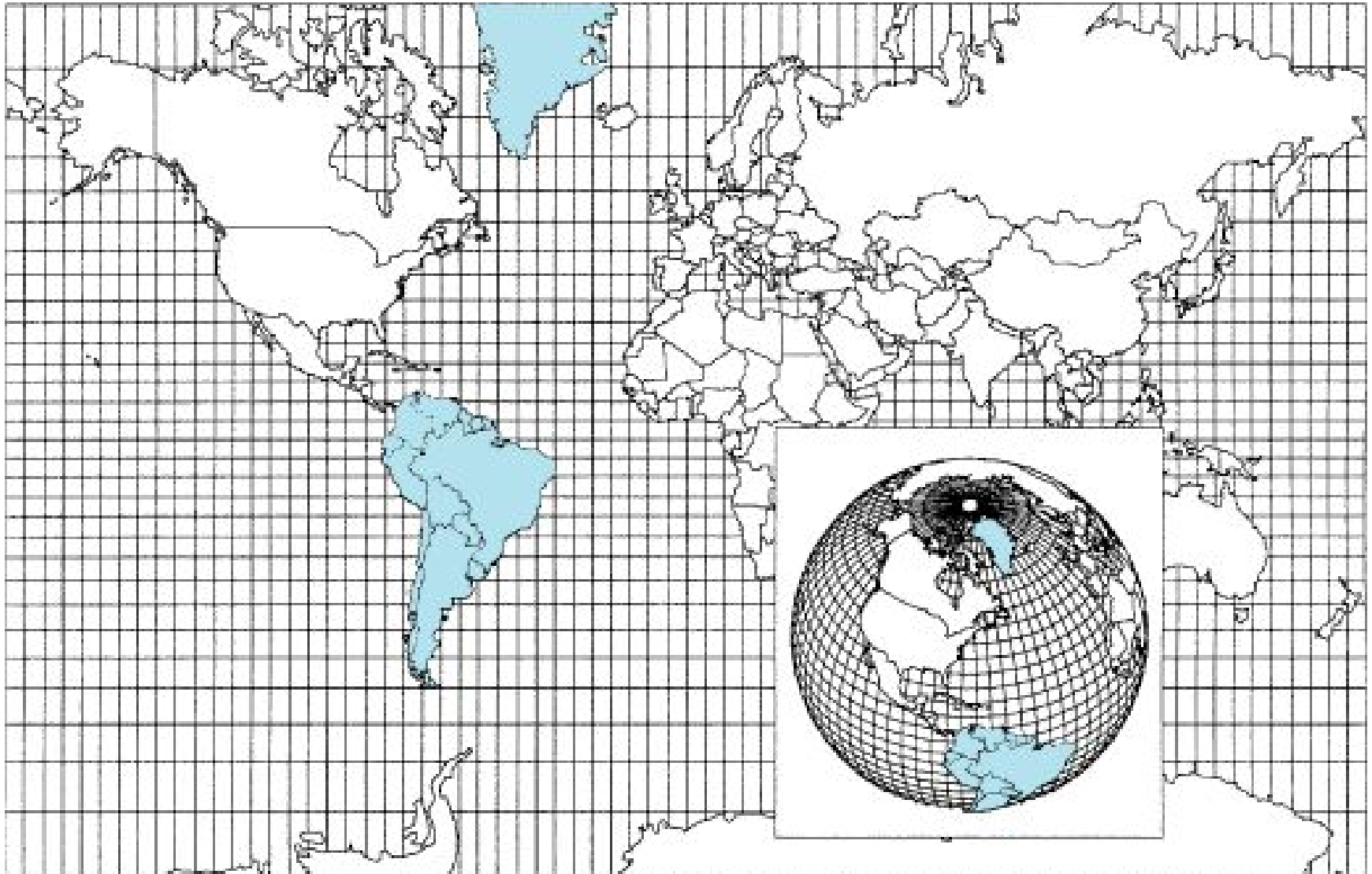
Mercator Projection

- Scale changes with latitude

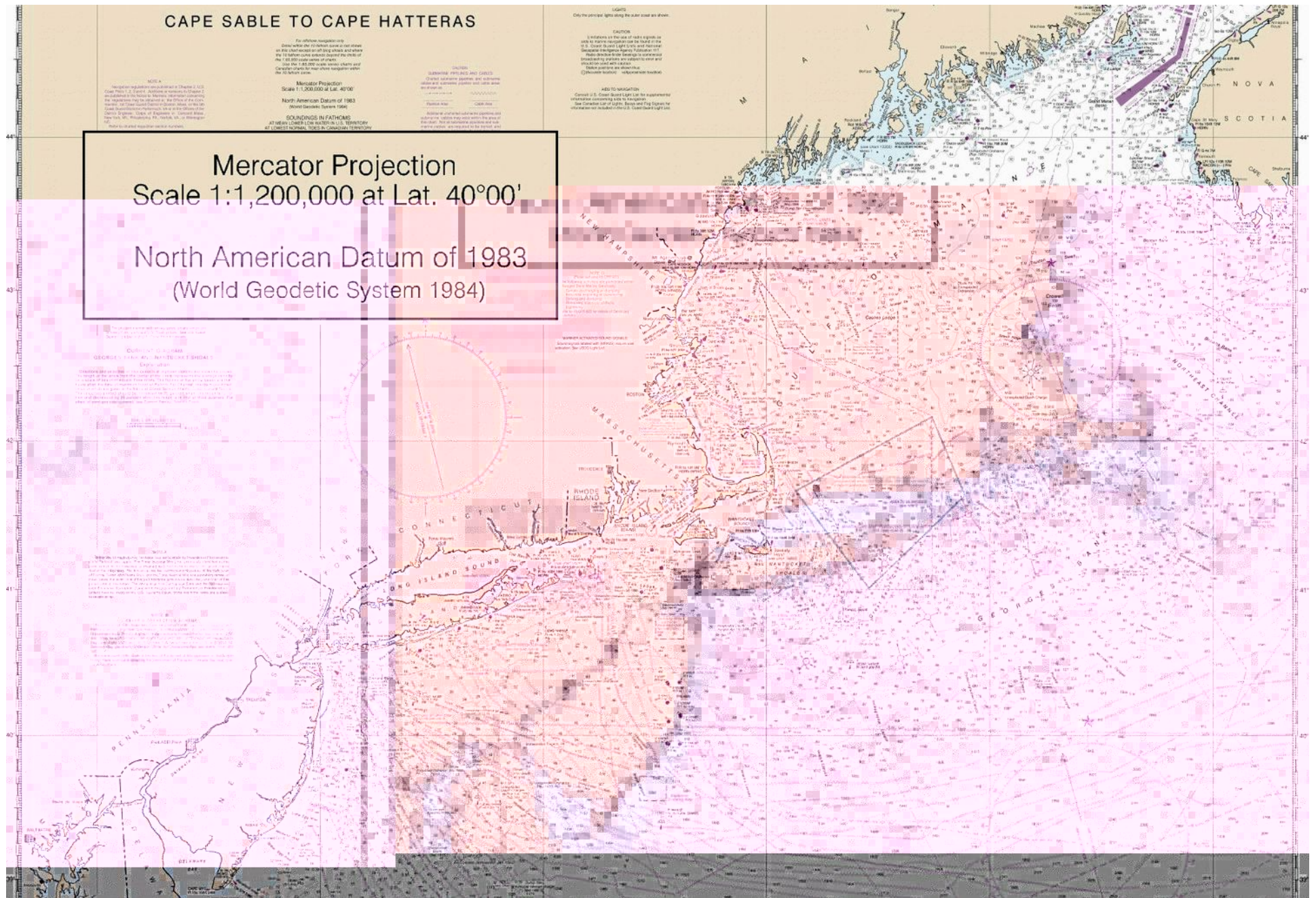


Mercator Projection

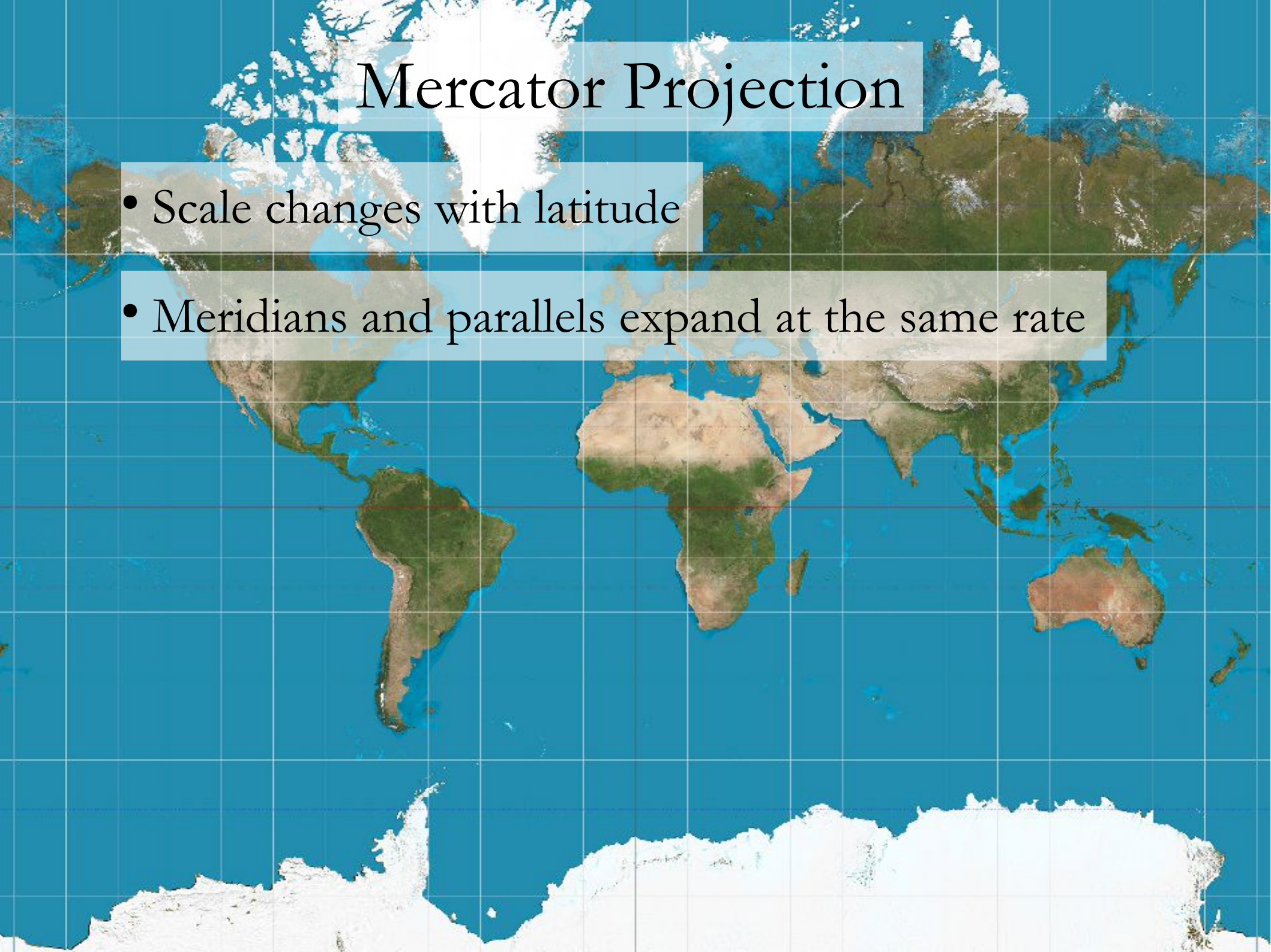
- Scale changes with latitude



Mercator Projection

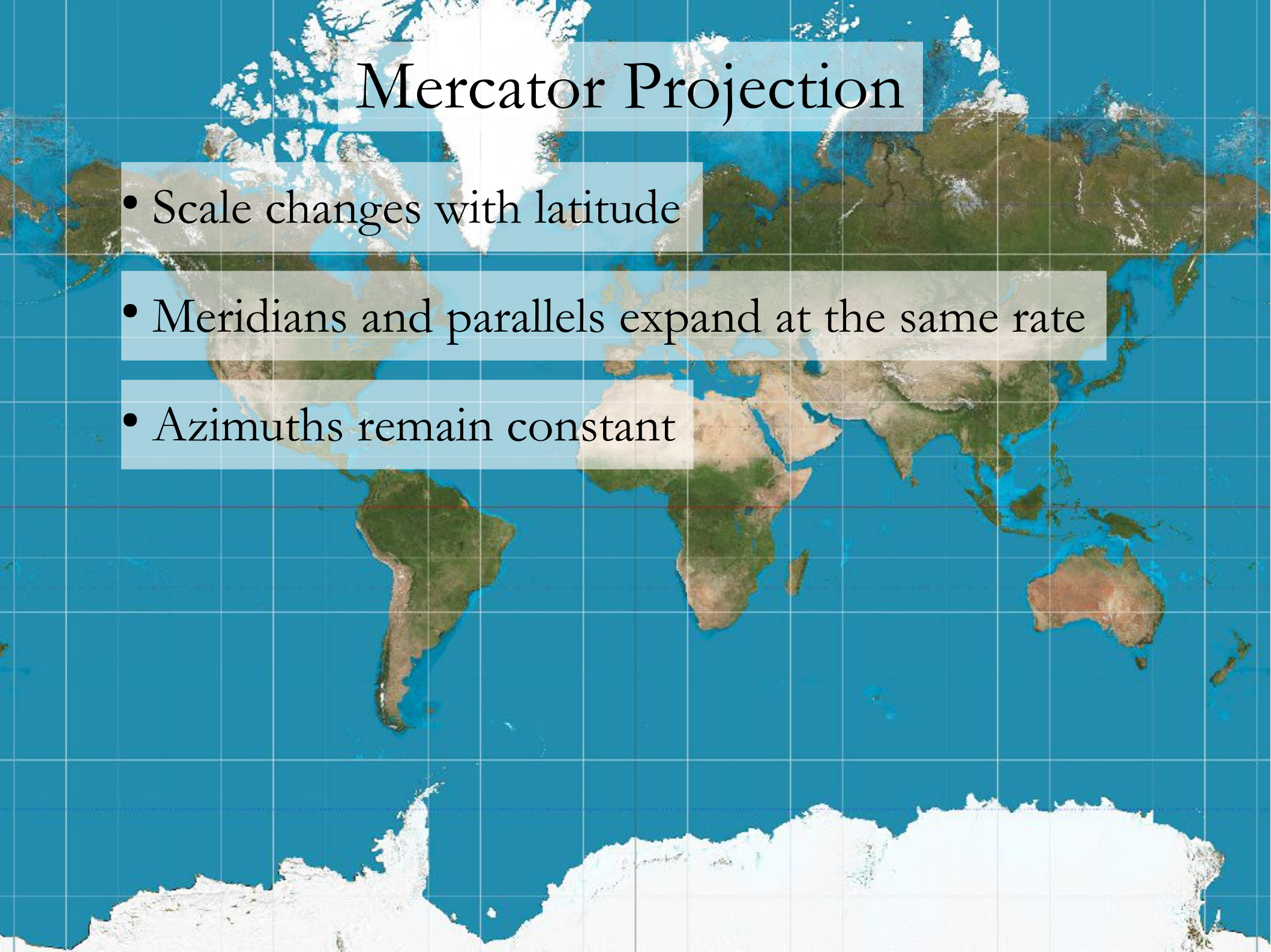


Mercator Projection



- Scale changes with latitude
- Meridians and parallels expand at the same rate

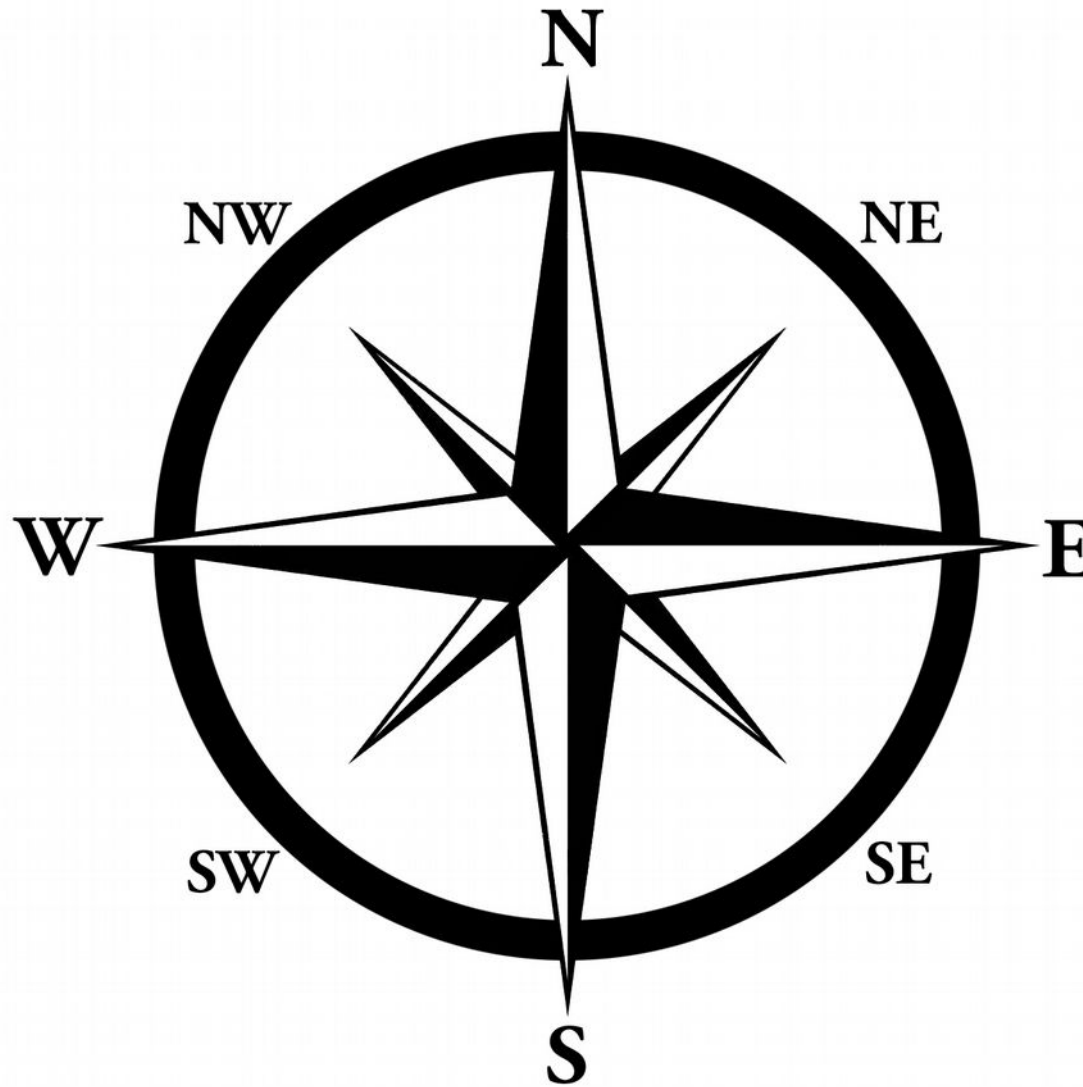
Mercator Projection



- Scale changes with latitude
- Meridians and parallels expand at the same rate
- Azimuths remain constant

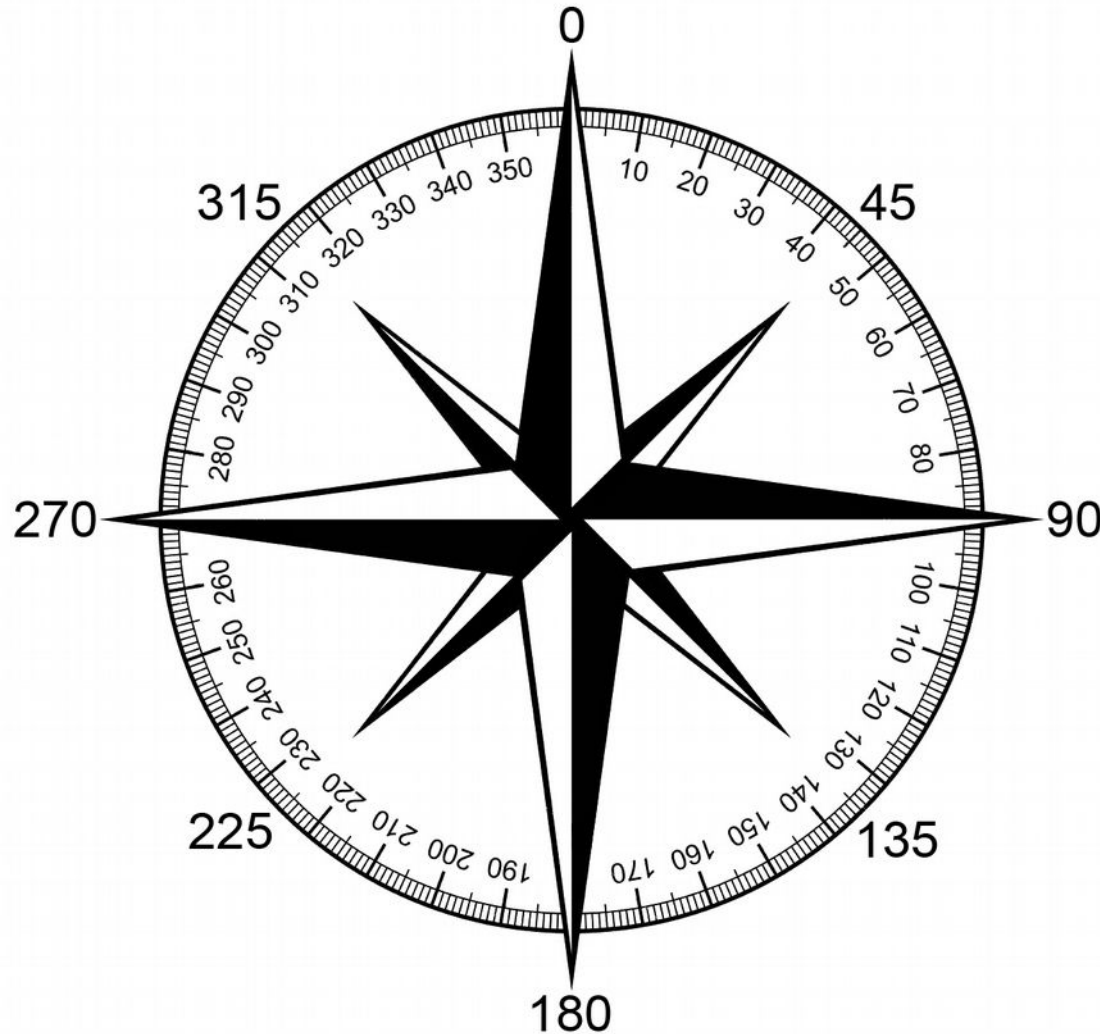
Azimuth

Azimuth is a direction or angle parallel to the horizon, usually in degrees, referenced to true north.

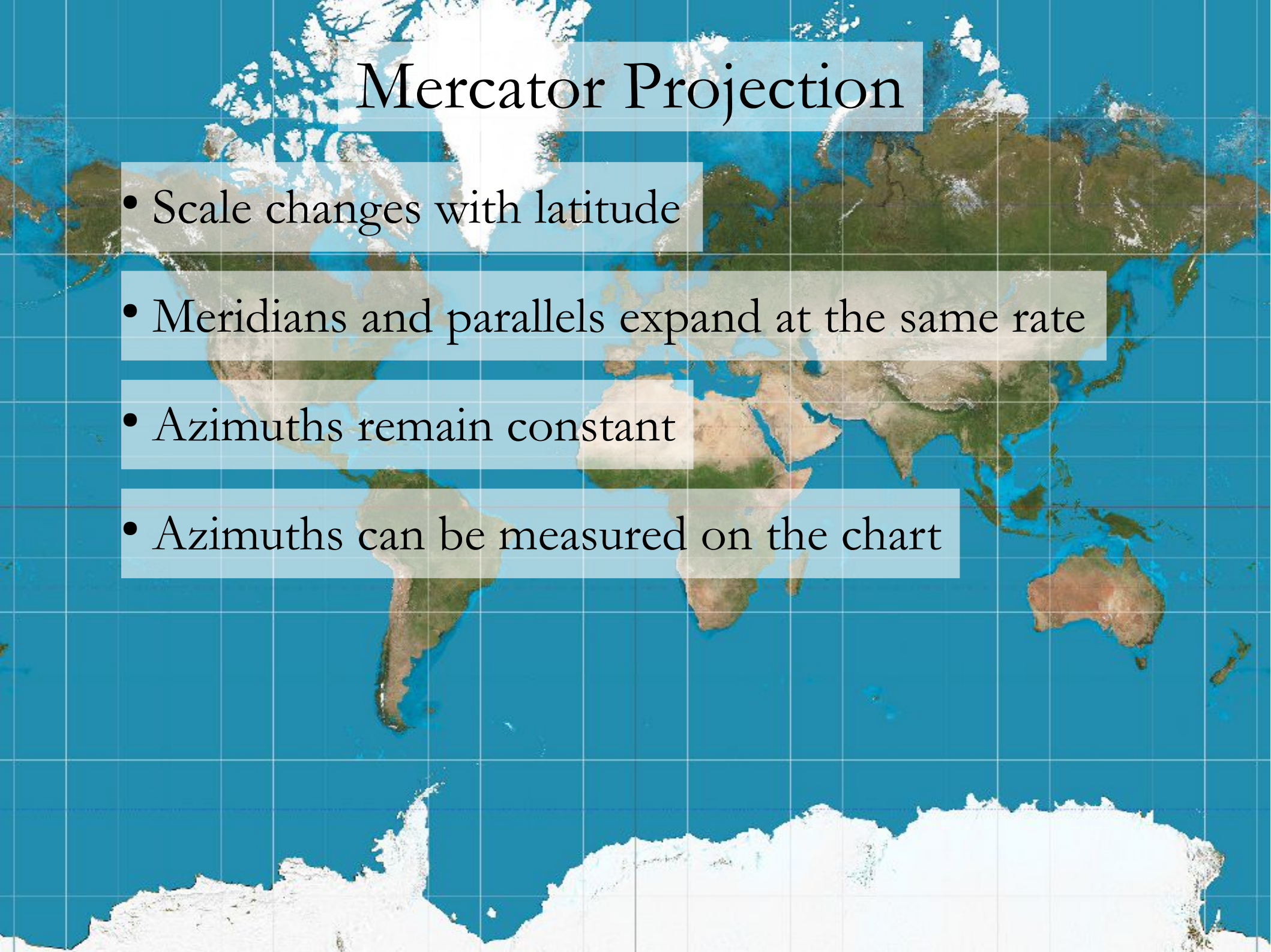


Azimuth

Azimuth is a direction or angle parallel to the horizon, usually in degrees, referenced to true north.

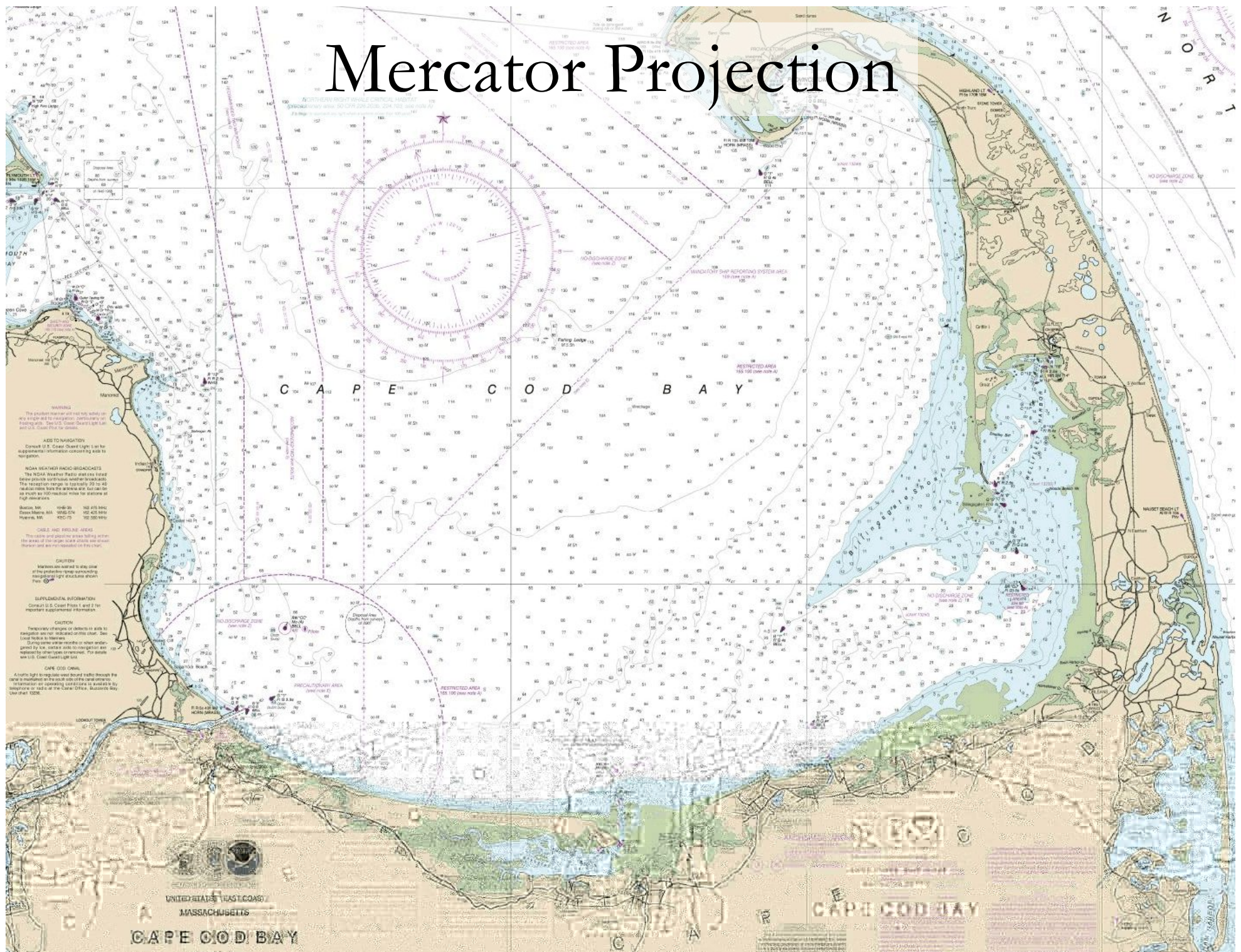


Mercator Projection

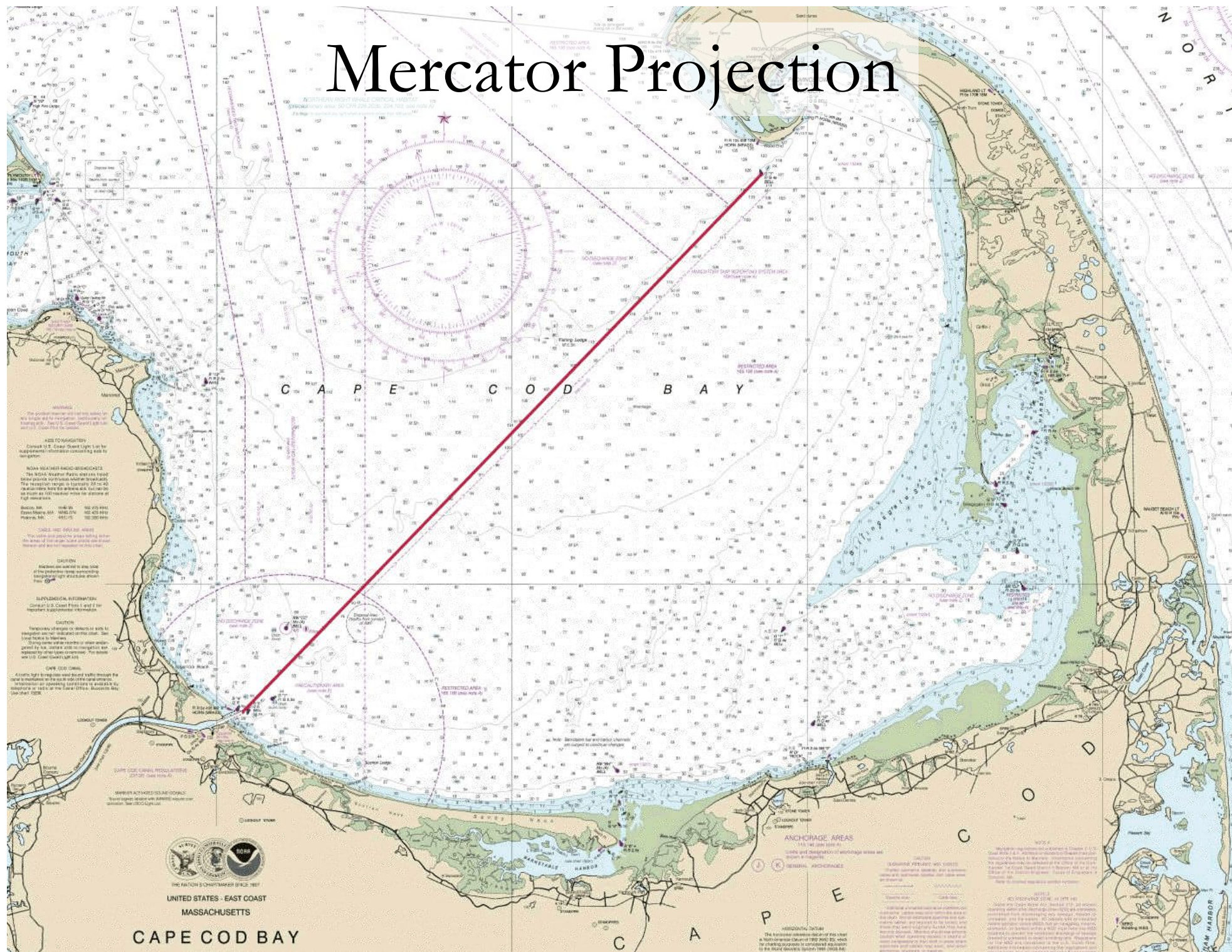


- Scale changes with latitude
- Meridians and parallels expand at the same rate
- Azimuths remain constant
- Azimuths can be measured on the chart

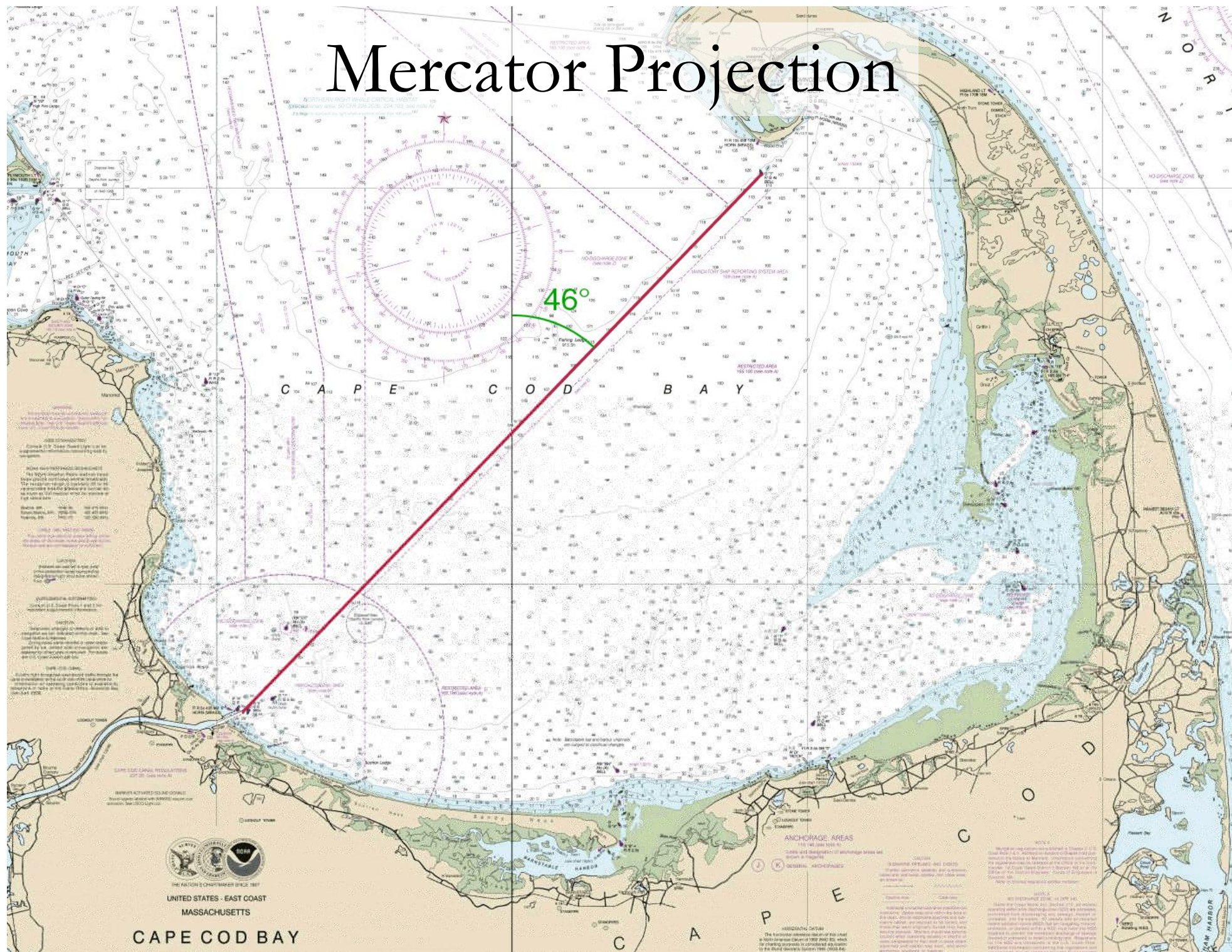
Mercator Projection



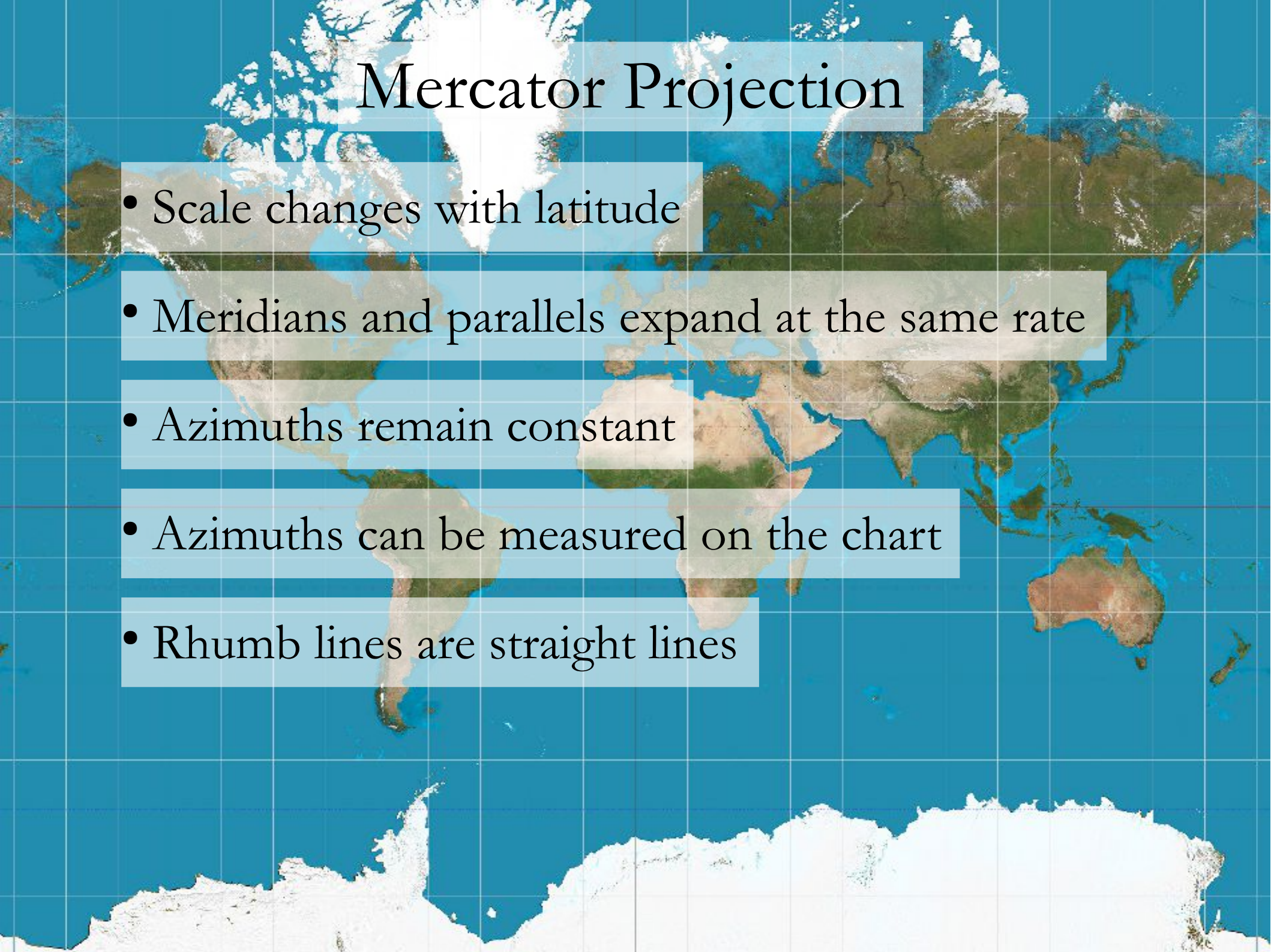
Mercator Projection



Mercator Projection



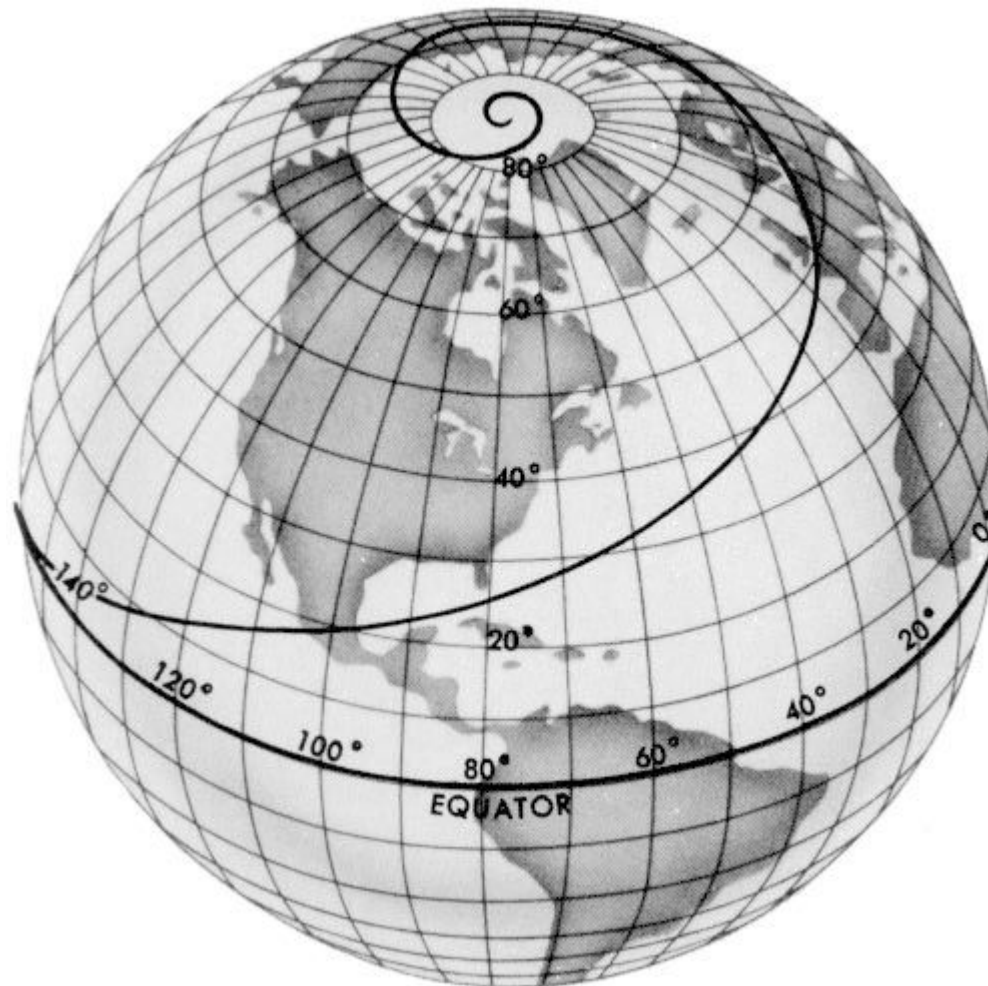
Mercator Projection



- Scale changes with latitude
- Meridians and parallels expand at the same rate
- Azimuths remain constant
- Azimuths can be measured on the chart
- Rhumb lines are straight lines

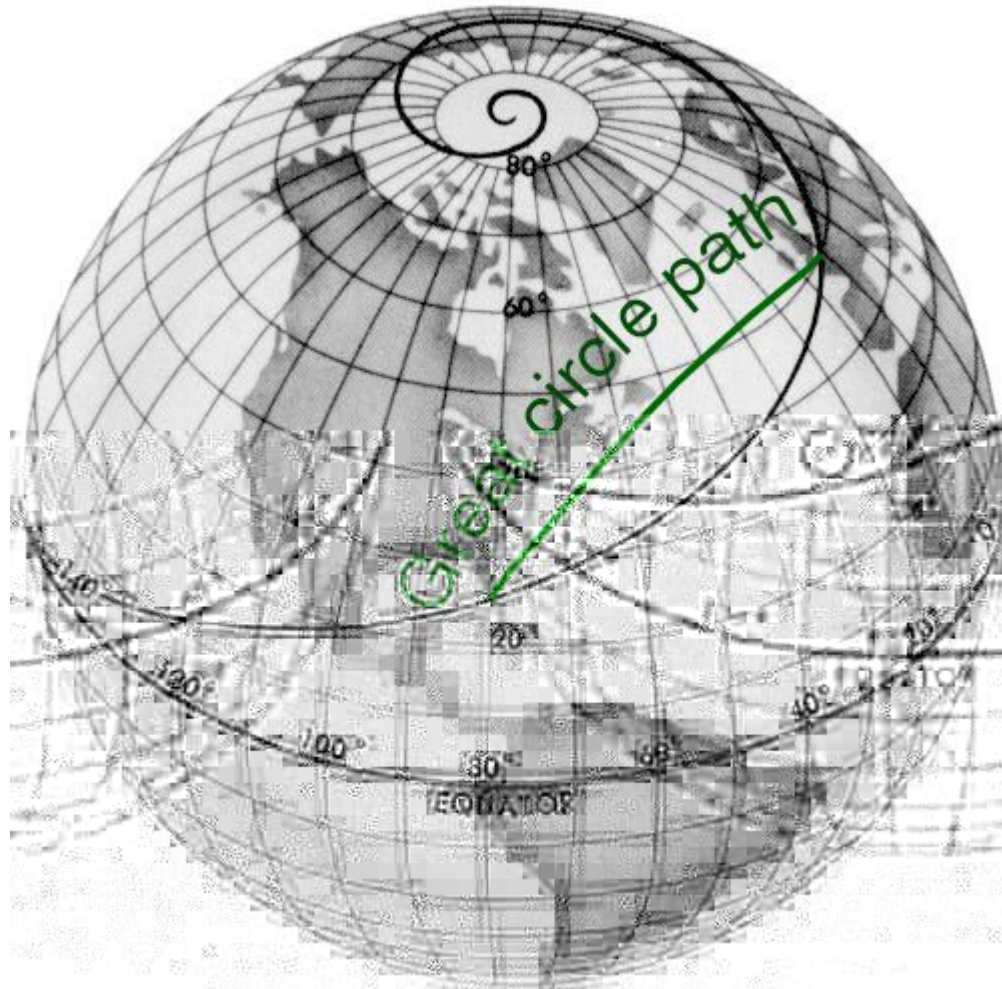
Mercator Projection

- A rhumb line is a path of constant azimuth.
(also called a loxodrome).

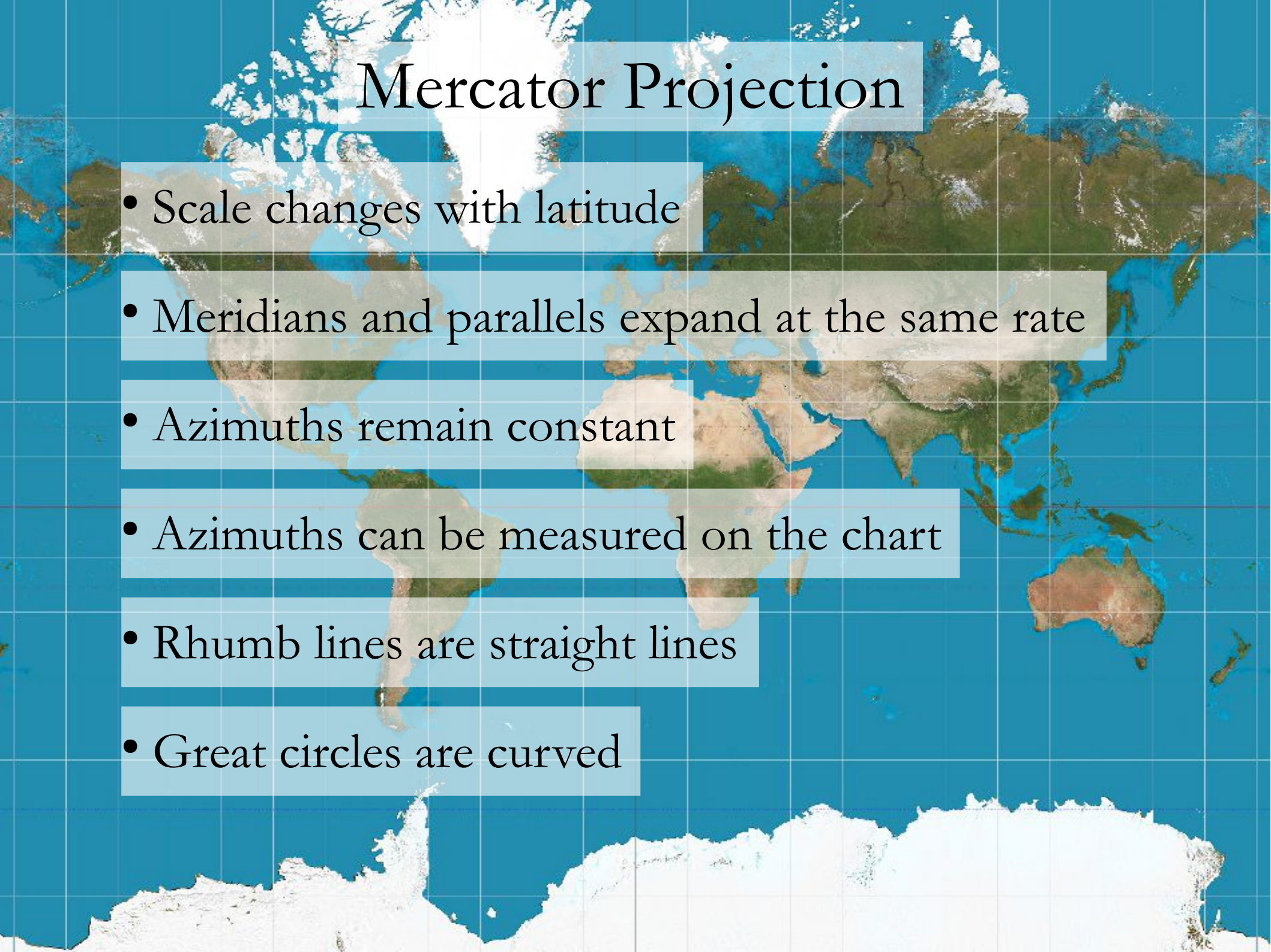


Mercator Projection

- The shortest distance between two points on a sphere is along a great circle.

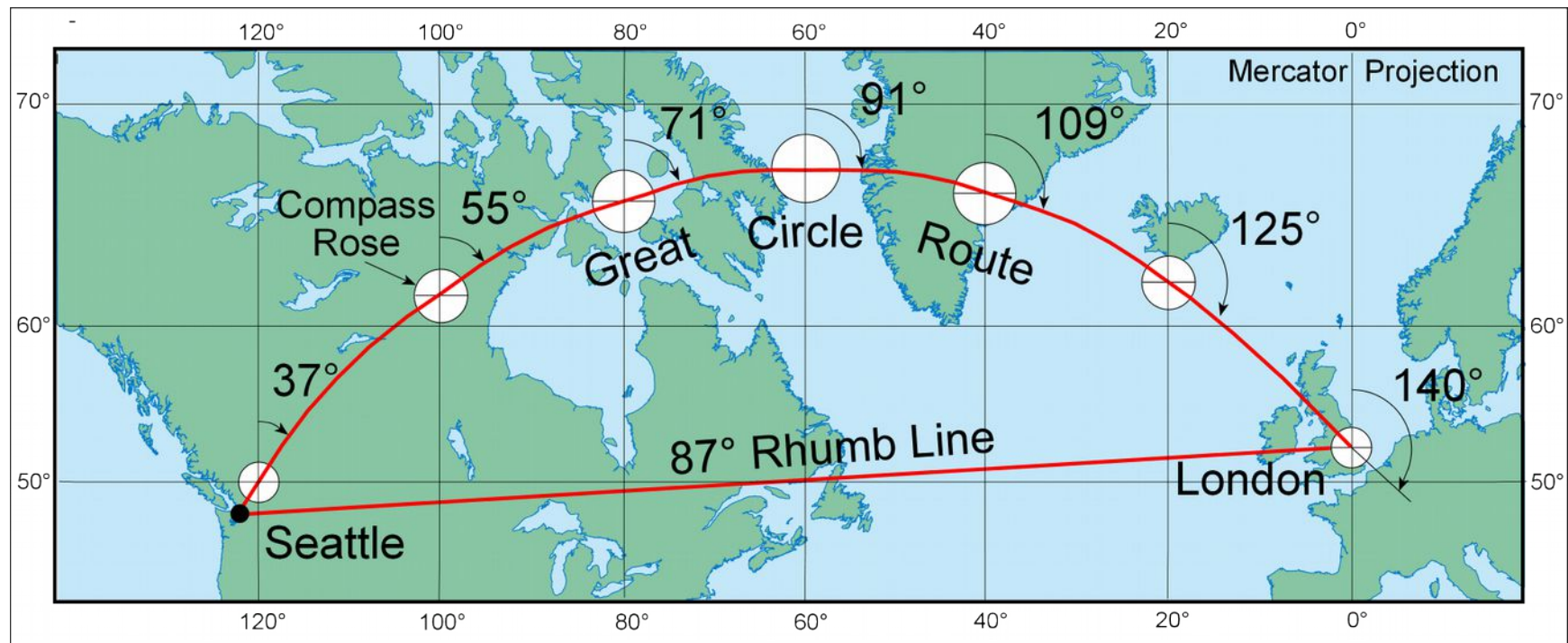
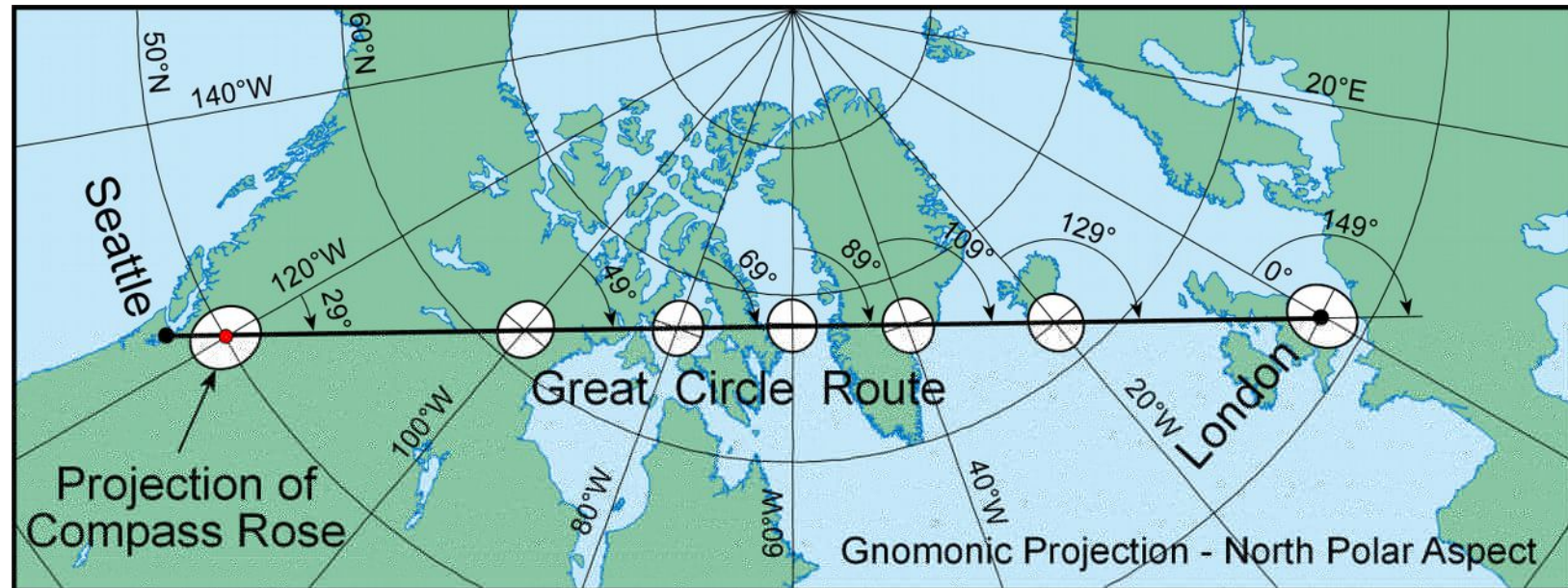


Mercator Projection



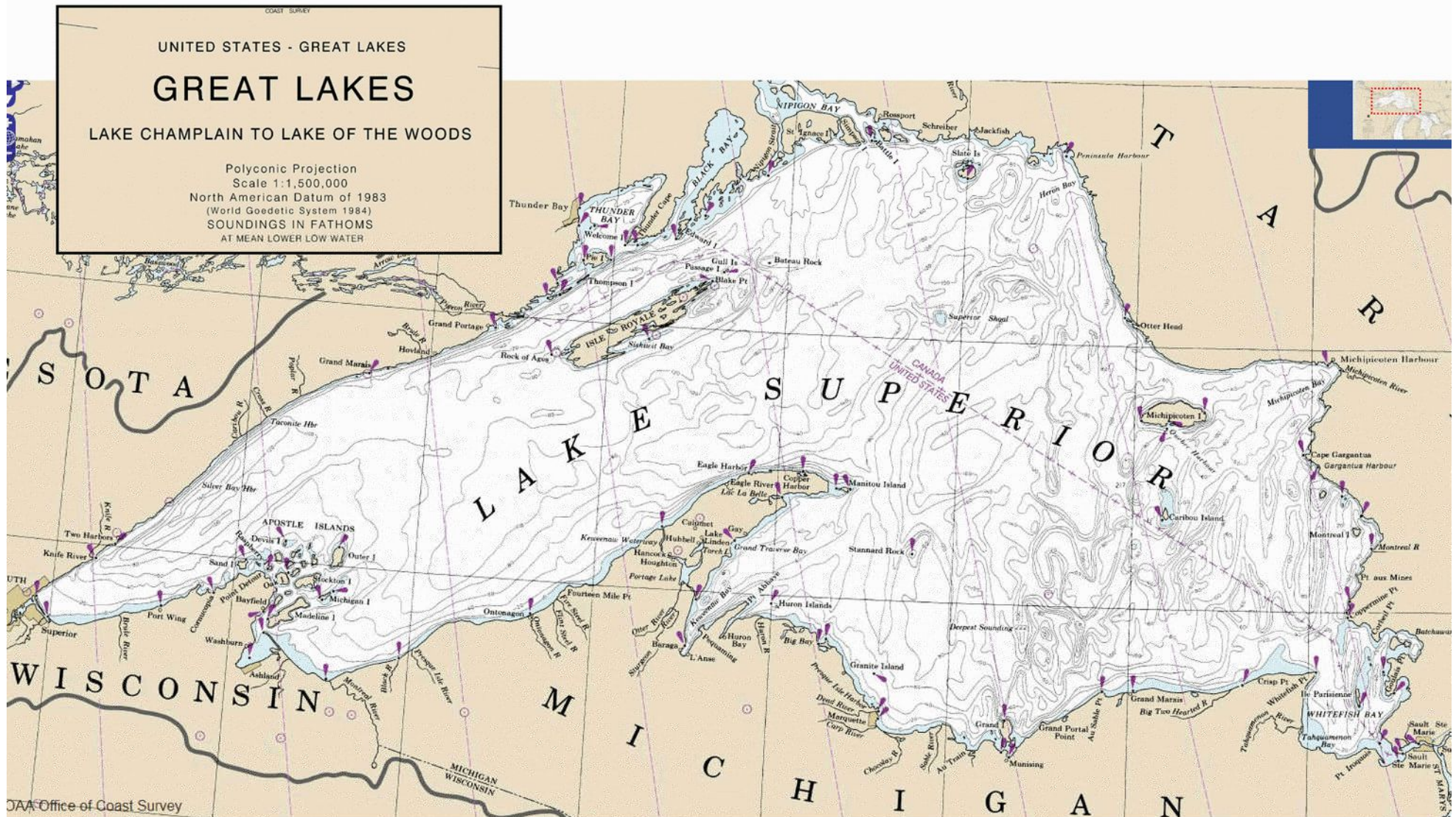
- Scale changes with latitude
- Meridians and parallels expand at the same rate
- Azimuths remain constant
- Azimuths can be measured on the chart
- Rhumb lines are straight lines
- Great circles are curved

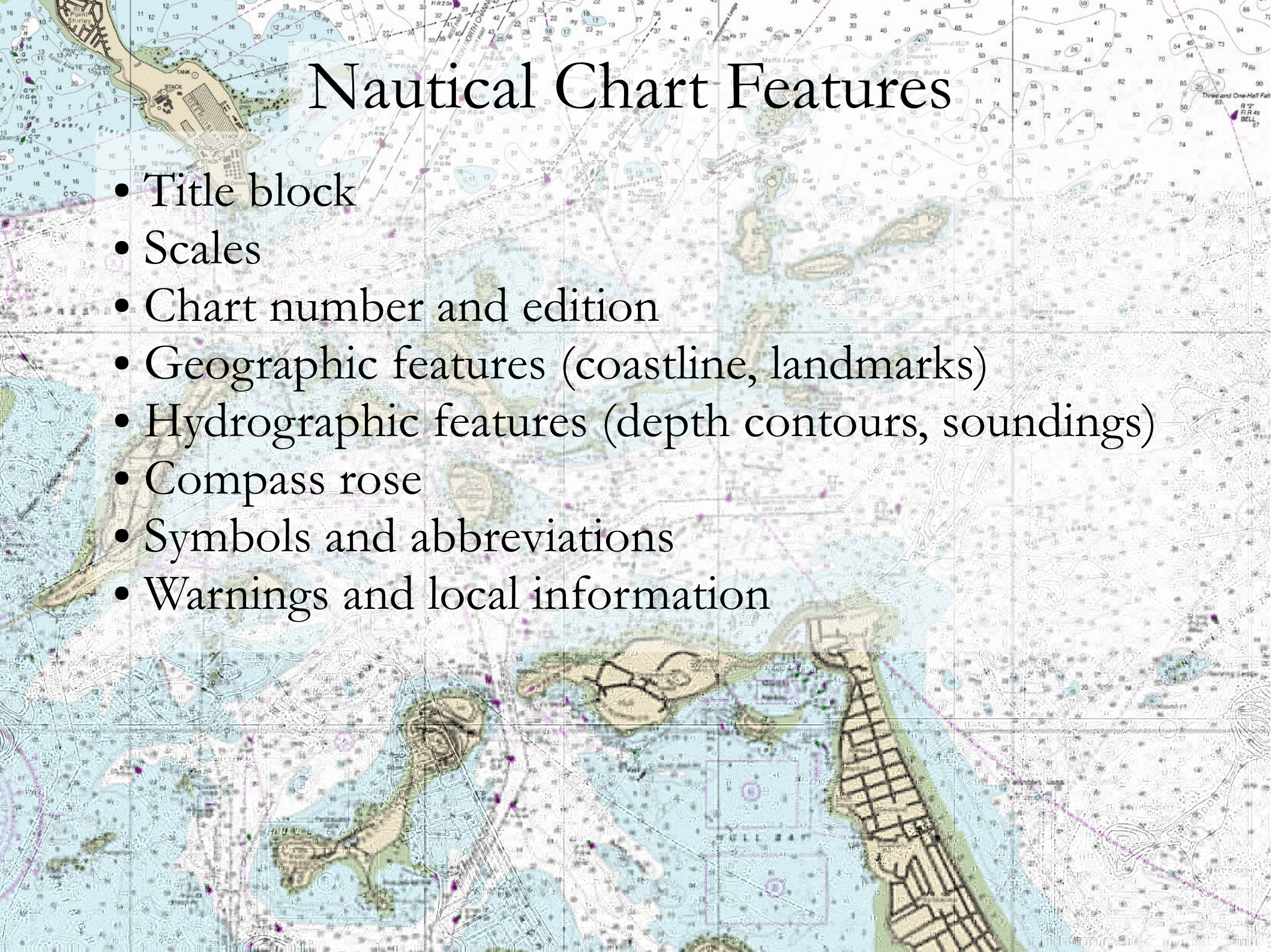
Mercator Projection



Polyconic Projection

Used on many charts of the Great Lakes





Nautical Chart Features

- Title block
- Scales
- Chart number and edition
- Geographic features (coastline, landmarks)
- Hydrographic features (depth contours, soundings)
- Compass rose
- Symbols and abbreviations
- Warnings and local information

Chart Title Block

UNITED STATES - EAST COAST

MASSACHUSETTS

BOSTON HARBOR

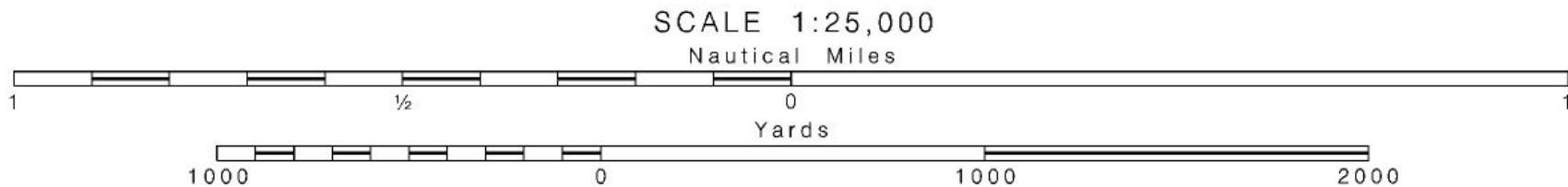
Mercator Projection
Scale 1:25,000 at Lat. 42°19'

North American Datum of 1983
(World Geodetic System 1984)

SOUNDINGS IN FEET
AT MEAN LOWER LOW WATER

Chart Scales

- Representative fraction (e.g. 1:80,000)
- A statement (e.g. “one inch equals 30 miles”)
- Graphic scale



- Latitude index along each side of the chart

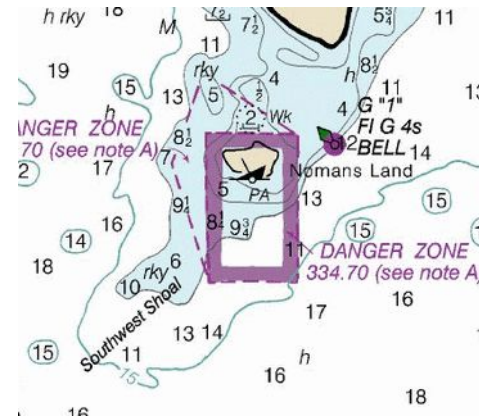
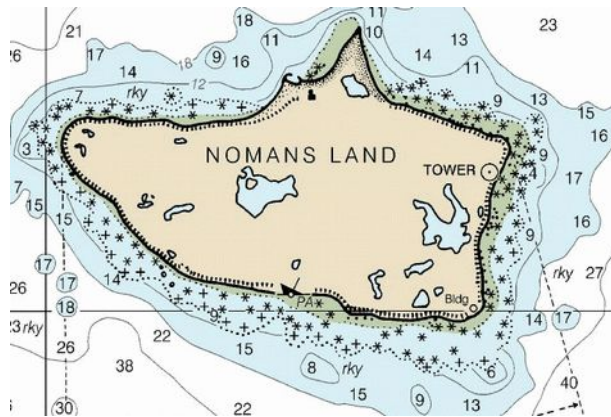
42°
30'

25'

Chart Scales

- “Large Scale” vs. “Small Scale”

Refers to the size of the printed images on the chart:



- 73,913 inches per nautical mile ($\approx 72,000$)

1:72,000 scale $\rightarrow$ 1 nm. $\approx$ 1 inch

1:25,000 scale $\rightarrow$ 1 nm. $\approx$ 3 inches

1:800,000 scale $\rightarrow$ 1 nm. $\approx$ 1/10 inch

Chart Scales

- **Sailing Charts**

very small scale – 1:600,000 or greater
(covers areas over 300 miles)

- **General Charts**

1:150,000 to 1:600,000
(75 – 300 miles)

- **Coastal Charts**

1:50,000 to 1:150,000
(25 – 75 miles)

- **Harbor Charts**

large scale – 1:50,000 or less
(less than 25 miles)

Chart Number and Edition

64th Ed., Feb. / 11

13270

CAUTION

This chart has been corrected from the Notice to Mariners (NM) published weekly by the National Geospatial-Intelligence Agency and the Local Notice to Mariners (LNM) issued periodically by each U.S. Coast Guard district to the dates shown in the lower left hand corner. Chart updates corrected from Notice to Mariners published after the dates shown in the lower left hand corner are available at nauticalcharts.noaa.gov.

Last Correction: 12/15/2015. Cleared through:
LNM: 5015 (12/15/2015), NM: 5215 (12/26/2015), CHS: 1115 (11/27/2015)

- Chart number: 5 digits
- Above number is edition number and print date
- Corrections to charts are published in Notices to Mariners (NM) and Local Notices to Mariners (LNM)
- New corrections should be applied to chart



U.S. Department
of Homeland Security
**United States
Coast Guard**

LOCAL NOTICE TO MARINERS

District: 1

Week: 52/15

COASTAL WATERS FROM EASTPORT, MAINE TO SHREWSBURY, NEW JERSEY

NOTES:

- (1) Unless otherwise indicated, missing and destroyed structures are presumed to be in the immediate vicinity of assigned position. Mariners should proceed with caution.
- (2) The Local Notice to Mariners is a weekly edition.
- (3) Inquiries, published articles or Information: mail to: LNM@uscg.mil
- (4) The U.S. Coast Pilot supplements the navigational information shown on nautical charts.
- (5) The Coast Pilot, along with its corrections, are available online at <http://www.nauticalcharts.noaa.gov/nsd/cpdownload.htm>.

The Local Notice to Mariners is available online at <http://www.navcen.uscg.gov/?pageName=lnmDistrict®ion=1>

The updated 2015 Light List is available online at: <http://www.navcen.uscg.gov/?pageName=lightListWeeklyUpdates>

Information on Private Aids to Navigation is available at: <http://www.uscg.mil/d1/prevention/NavInfo/navinfo/paton.htm>

Reports of Channel conditions can be found at the Army Corps of Engineers website at:

- NM is for large ships, LNM is for all boats
- Published weekly, number refers to week/year
- Subscribe or download at <http://navcenter.uscg.gov>

Chart Geographic Features

- Land: tan
- Intertidal Zone: green
- Water: white
- Shallow Water: blue

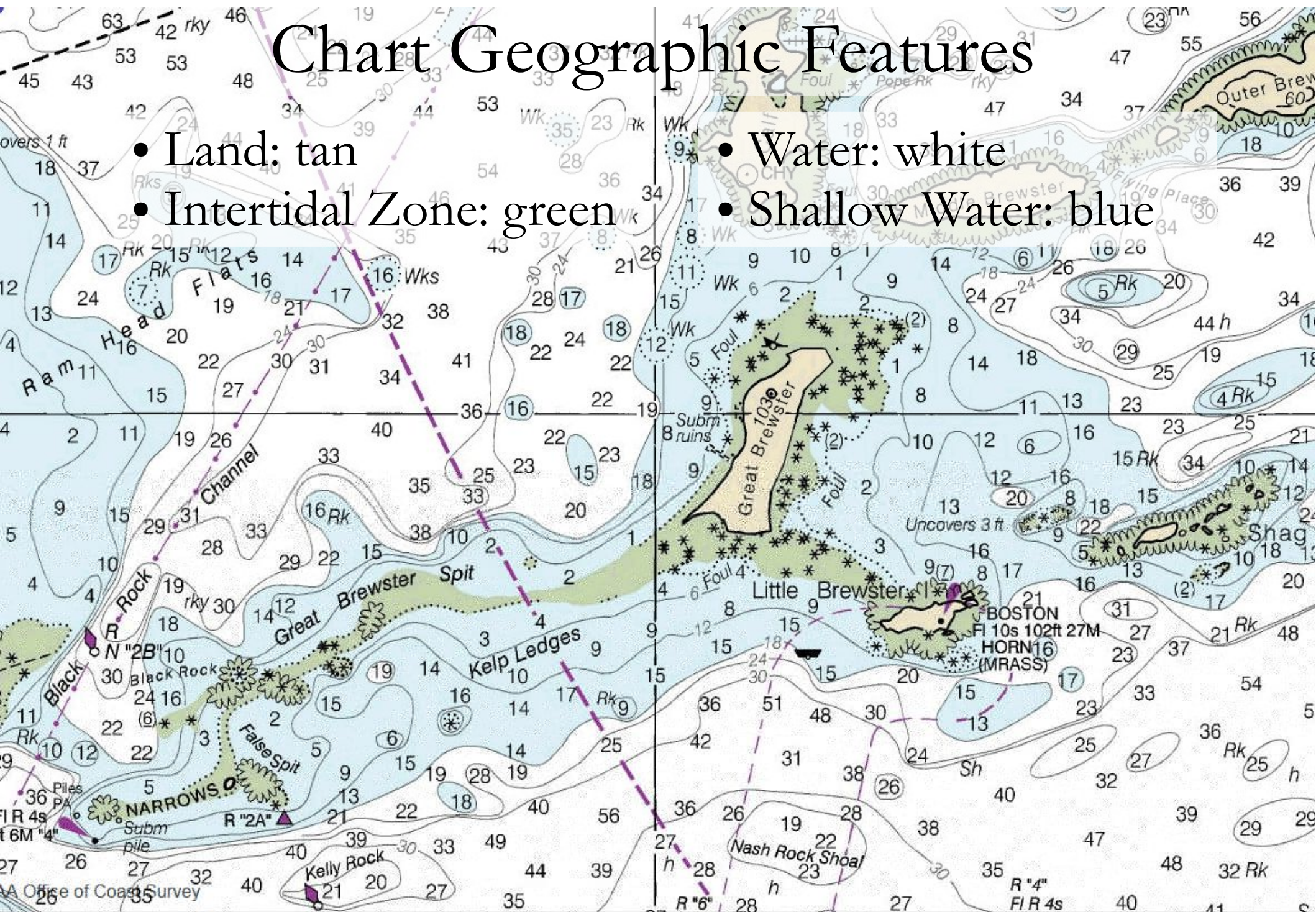


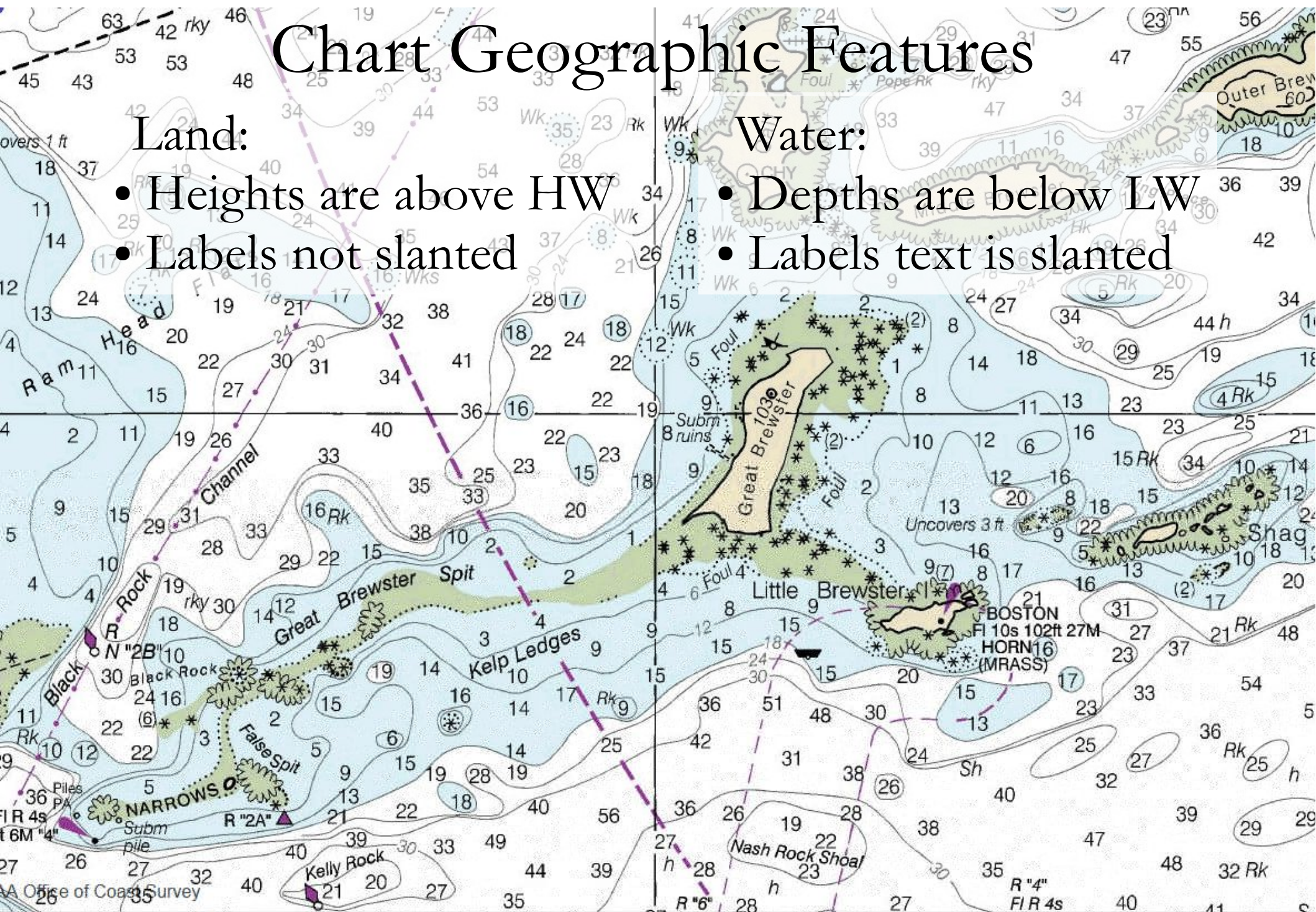
Chart Geographic Features

Land:

- Heights are above HW
- Labels not slanted

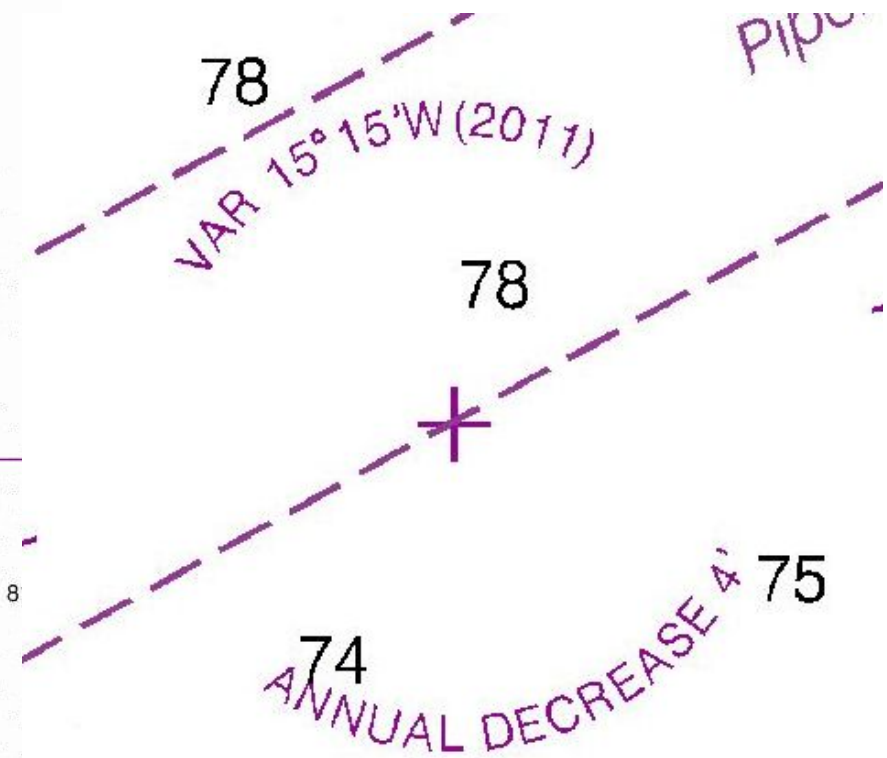
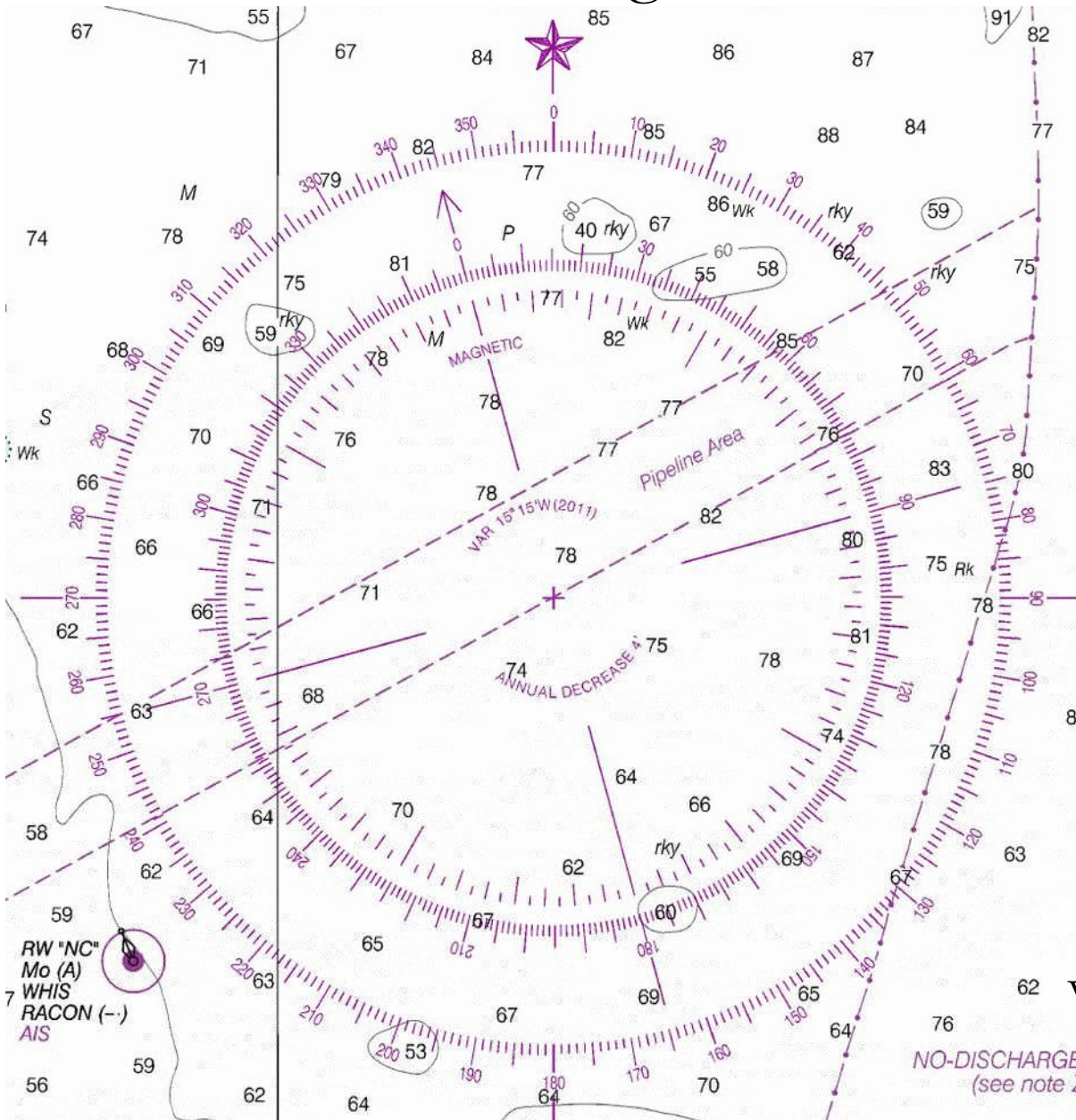
Water:

- Depths are below LW
- Labels text is slanted



Compass Rose

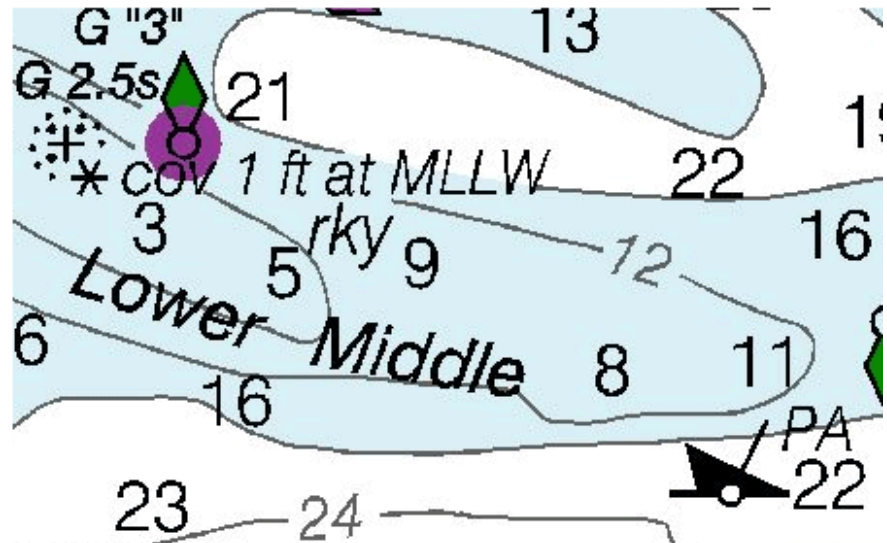
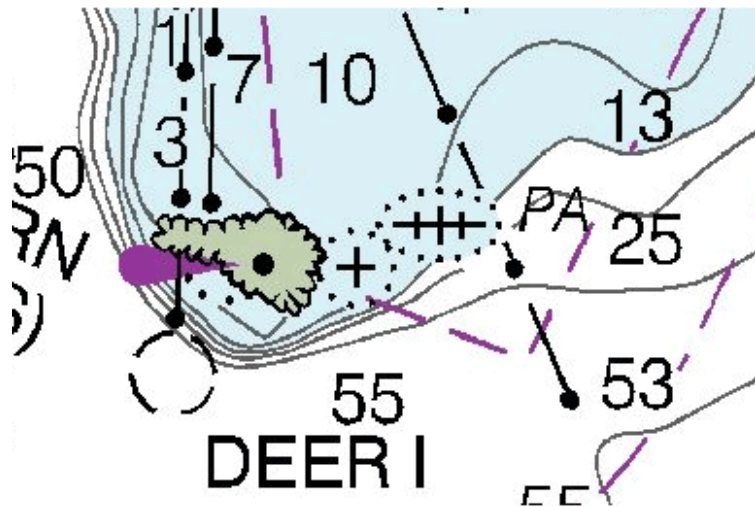
Outside ring: True, Inside ring: Magnetic



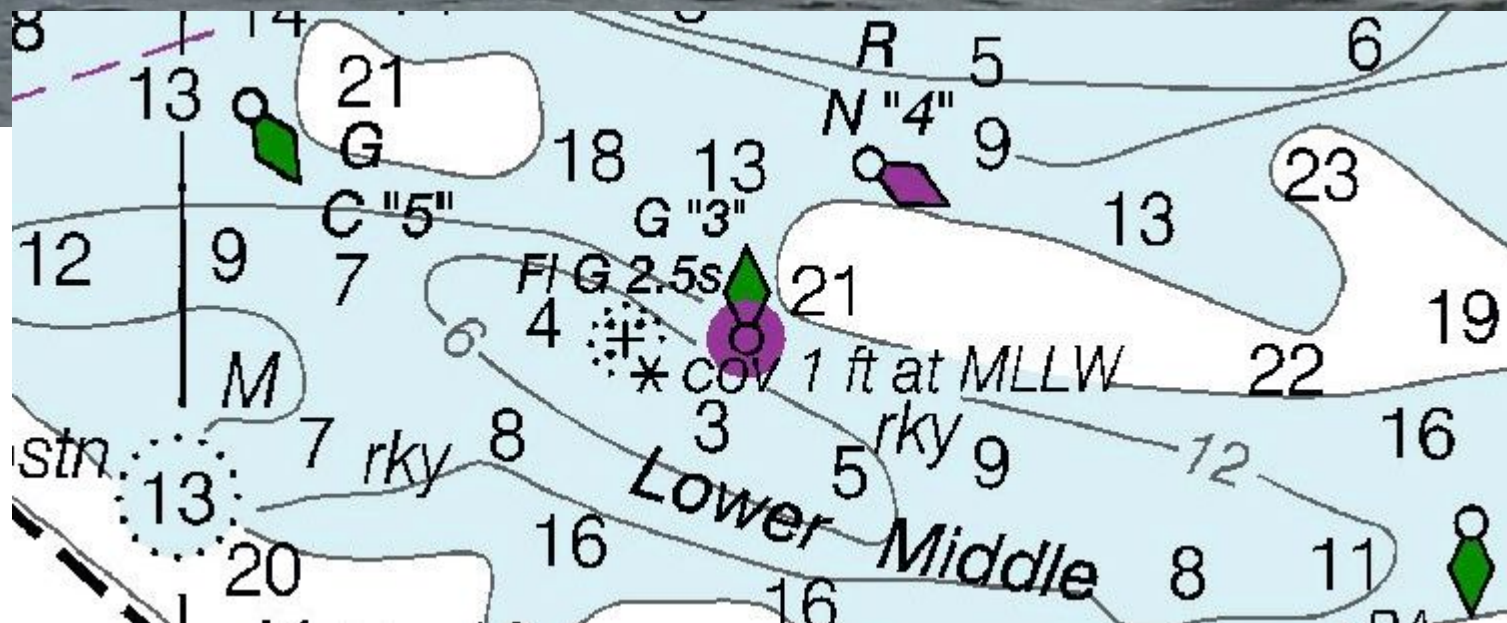
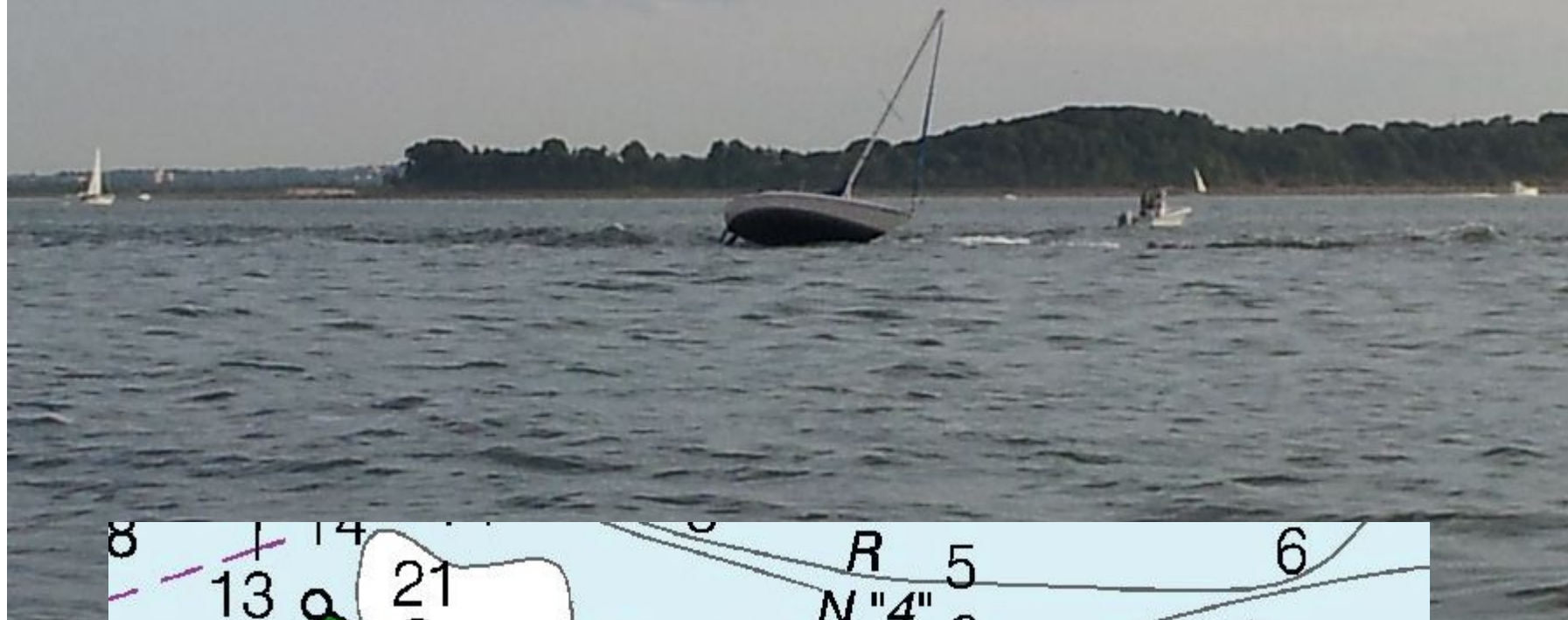
Increase or decrease of absolute value

Chart Symbols

- + Rock (below water at low tide)
- ⦿ Rock (at low water level)
- * Rock (above water at low tide)
- ⚓ Wreck (above water at low tide)
- ++ Wreck (below water at low tide)
-  Bouy
-  Lighted Bouy
-  Light



Lower Middle Channel



All chart symbols are listed in “Chart No. 1”

Rocks, Wrecks, Obstructions, Aquaculture K

No.	INT	Description	NOAA	NGA	Other NGA	ECDIS
General						
1		Danger line: A danger line draws attention to a danger which would not stand out clearly enough if represented solely by its symbol (e.g. isolated rock) or delimits an area containing numerous dangers, through which it is unsafe to navigate				 Obstruction, depth not stated Obstruction which covers and uncovers Underwater hazard with depth of 20 meters or less Isolated danger of depth less than the safety contour  Foul area, not safe for navigation

Rocks		Plane of Reference for Heights = $\pm H$	Plane of Reference for Depths = $\pm H$	
10		Rock (1) which does not cover: height above; height datum		
11		Rock (1) which covers an uncovered area: height above; height datum		
12		Rock (1) which covers an uncovered area: height above; height datum		

Chart Warnings & Local Information

SMALL CRAFT WARNINGS

Year round small-craft warnings will be displayed during daytime only on Metropolitan District Commission Police Patrol Boats underway in Inner Boston Harbor from Nantasket Beach (42° 16.2' N, 70° 51.5' W) to waters around Georges and Lovell Islands.

NOTE B

PRECAUTIONARY AREA

Traffic within the Precautionary Area may consist of vessels operating between Boston Harbor and one of the established traffic lanes. Mariners are advised to exercise extreme care in navigating within this area.

Recommended traffic lanes have been established for the approach to Boston Harbor. Use charts 13200 and 13267.

Aids to Navigation (ATONs)



ATONs

- Bouys – floating, anchored to bottom
- Beacons – fixed to land, lit or unlit
 - Lights – fixed to land and lit
 - Daybeacons – fixed to land and unlit

ATON Identification

Ways to identify bouys:

- Floating (bouys) or fixed to land (beacons)
- Color (red, green, yellow, etc.)
- Shape (cylinder, cone, tower, ball, etc.)
- Topmark (ball, cone)
- Light color and pattern
- Numbers or letters
- Sound (bell, gong, whistle, fog horn)
- Radar transponder (RACON)

ATON Usage

- Lateral marks – mark sides of a channel (red, green)
- Center channel markers (red/white striped)
- Danger marks (red/black or other)
- Cardinal indicators (indicate safe water in one direction)
- Warnings/restrictions (white/orange)
- Other special purposes (yellow)

Lateral Marks

“Red Right Returning”

Usually mark a channel. Can be bouys, lights or daybeacons. Keep red lateral marks to starboard (to the right) when “returning” to a smaller harbor from a larger body of water.

Green:

- Odd Numbers
- Square or Cylinder

Red:

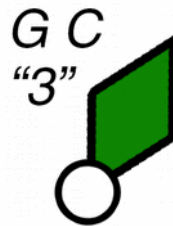
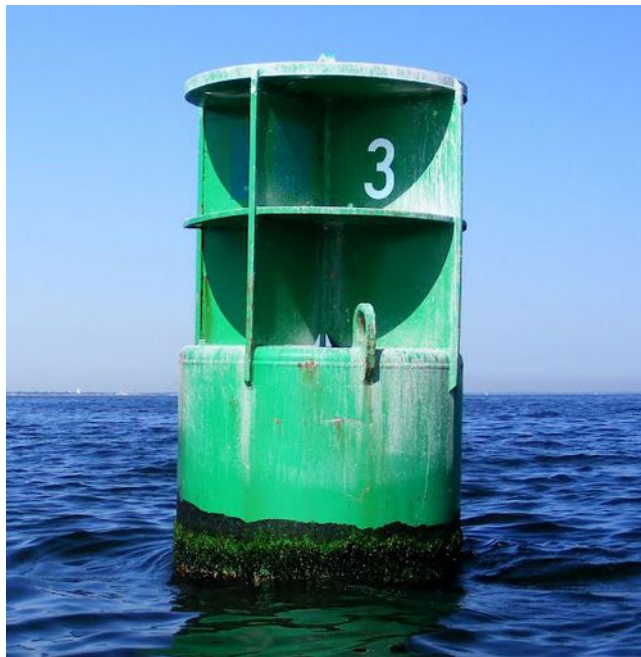
- Even Numbers
- Triangle or Cone

Cans & Nuns

- Are never lighted.
- Never have sounds
- When used as lateral marks:

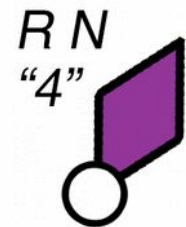
Cans:

- Green
- Odd number



Nuns:

- Red
- Even number

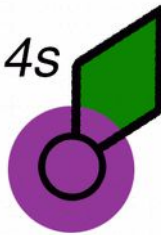


Tower Bouys

- May be lighted or have sounds.
- Light color usually matches bouy color



G "1"
Fl G 4s



R "20"
Q R

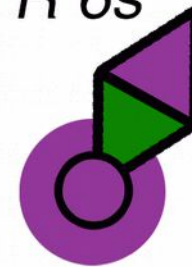


Preferred Channel Bouys

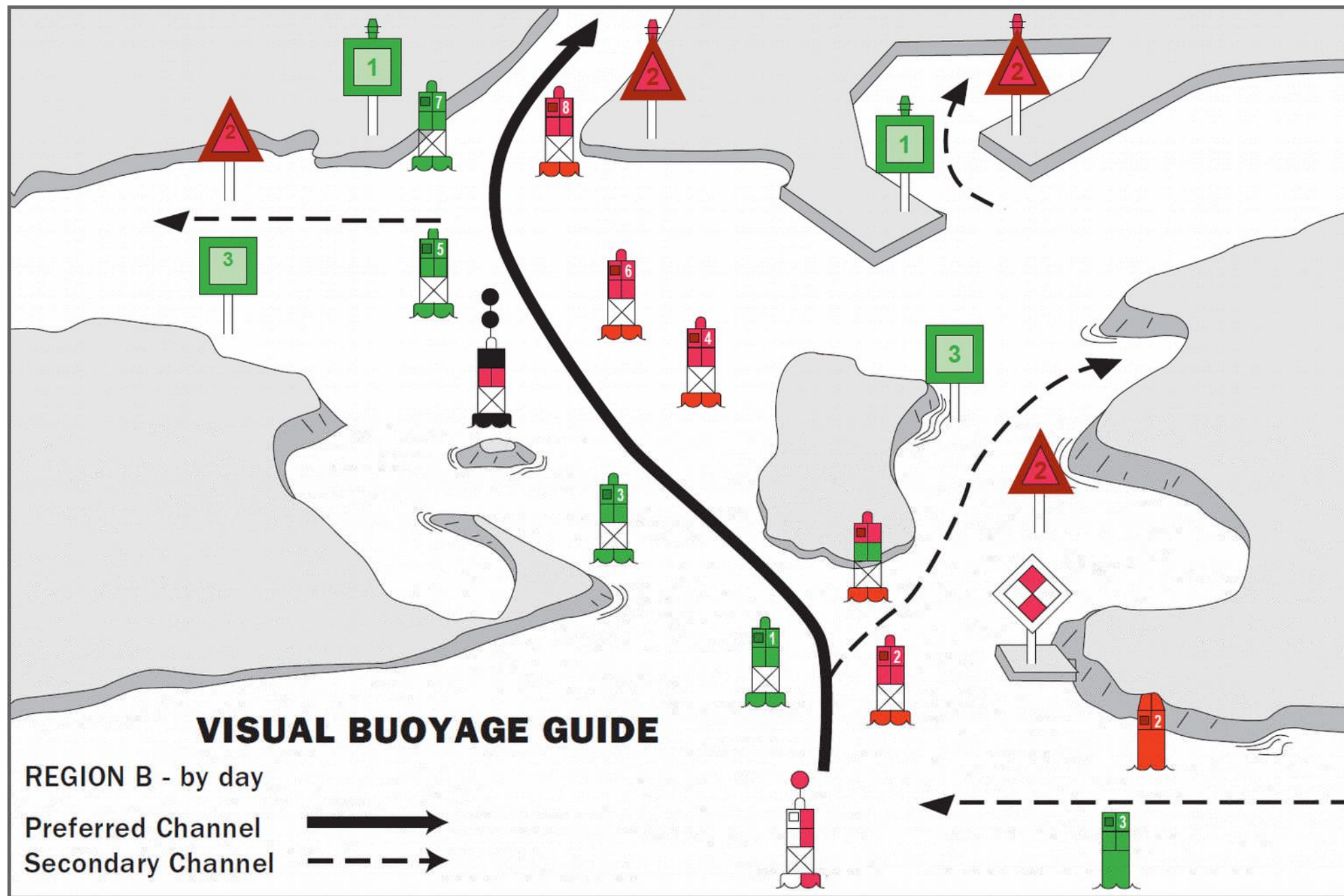
- Red/Green/Red or Green/Red/Green
- Placed at channel intersections
- Top color indicates preferred channel
- No numbers, may be lettered



RG "PR"
Fl (2+1) R 6s

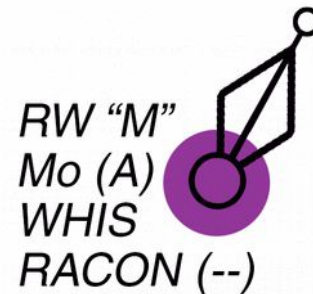


Lateral System



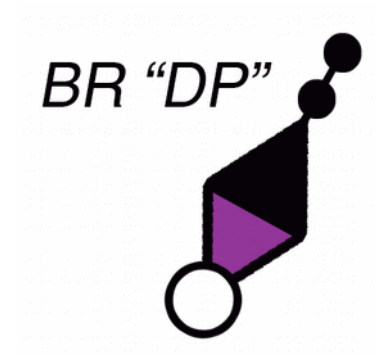
Center Channel Bouys

- Red & white vertical stripes
- Ball topmark
- If lit, white morse-A light pattern (•—)
- Sometimes have whistles
- Sometimes have radar transponders

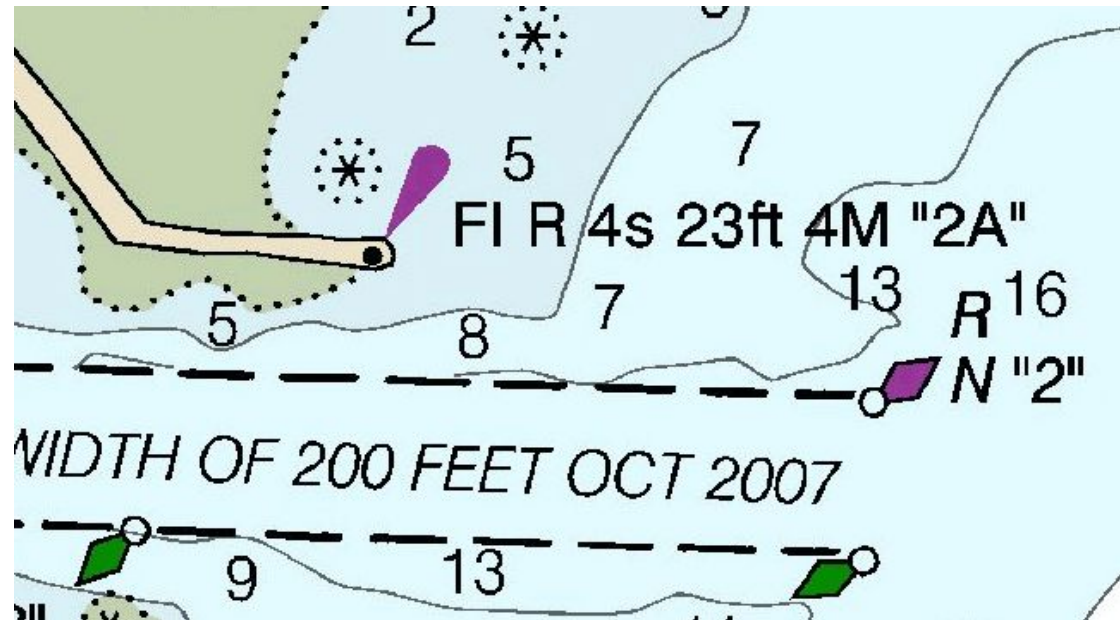


Danger Bouys

- Red & black
- Two black ball topmarks
- No numbers, may be lettered
- If lit, white (2) group flashing pattern



Lights



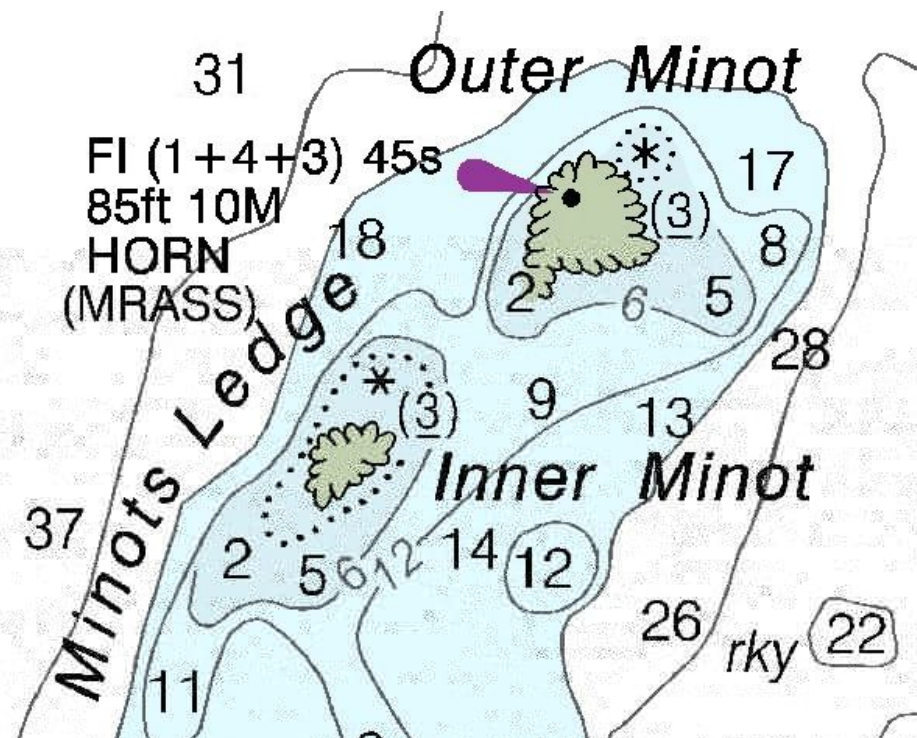
Scituate North Jetty Light 2A

Flashing Red 4s

23 feet above high water

4 miles nominal visibility

Lights

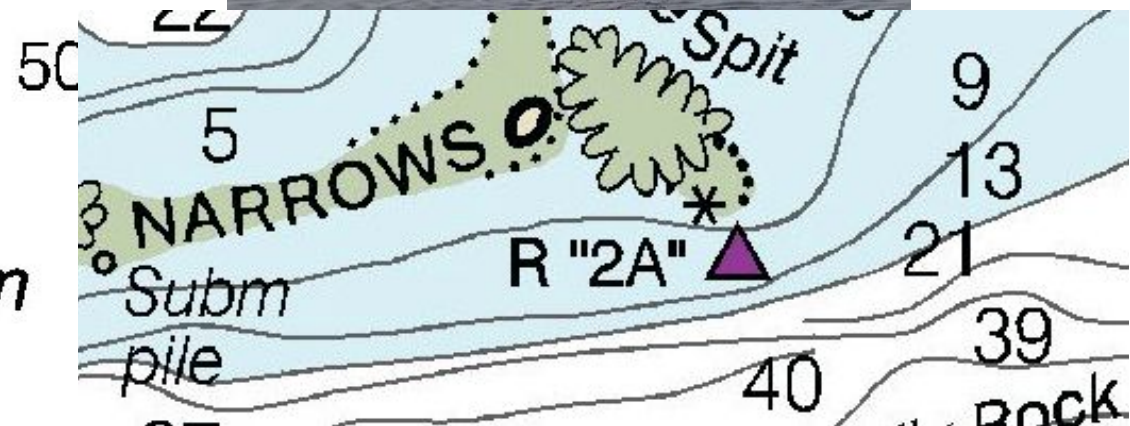
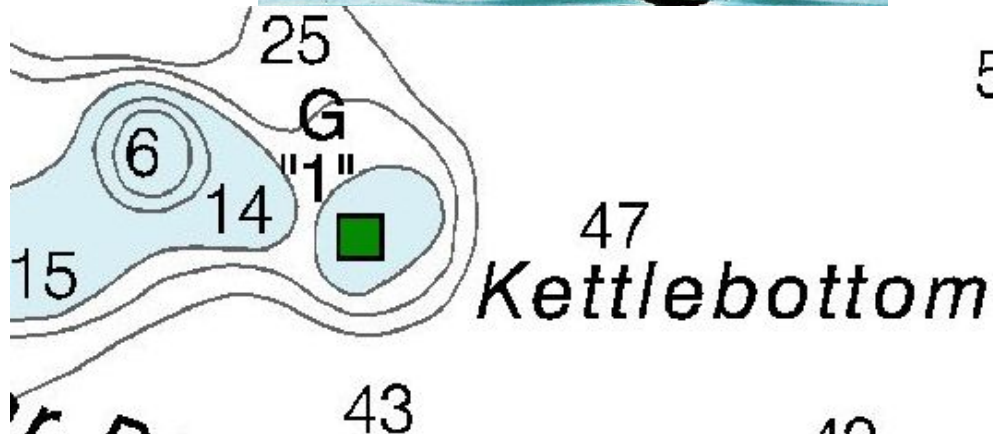


Minot Light

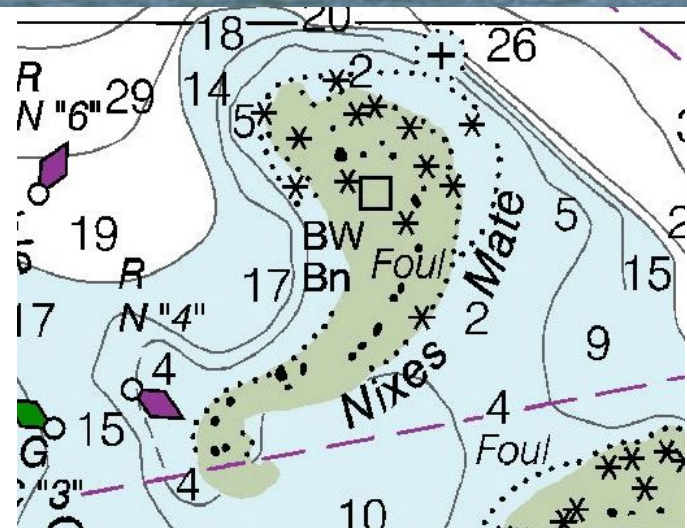
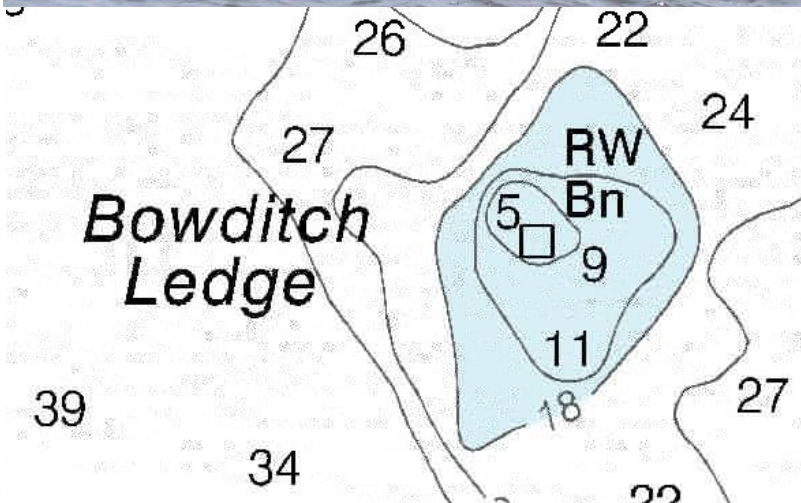
Group flashing (1+4+3) 45 sec.
85 feet above high water
10 miles nominal visibility
Fog Horn (MRASS)

Daybeacons

- No lights and fixed to land.



Daybeacons



Do Not Tie Up to Navigational Aids



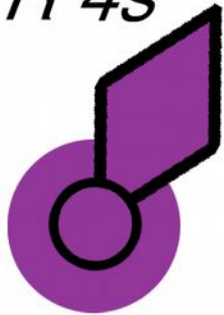
Light Characteristics

Description	Characteristic	Chart Abbreviation
Flashing		Fl
Quick flashing		Q
Group flashing		Fl (2)
Composite Group flashing		Fl (2+1)
Occulting		Oc
Group occulting		Oc (3)
Fixed		F
Isophase		Iso
Morse		Mo (letter)
Alternating		Al RWG

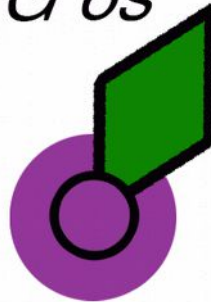
Light Characteristics

Group 1

R "6"
Fl R 4s



G "3"
Fl G 6s

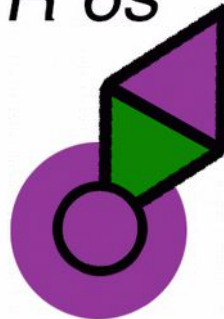


Y "B"
Fl Y 2.5s



Fl 10s 102ft 27M

RG "PR"
Fl (2+1) R 6s



Fl (4) 15s 35ft 12M

G "20"
Q G

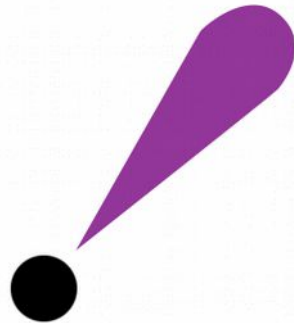
Fl (2) 5s 65ft 10M

Light Characteristics

Group 2



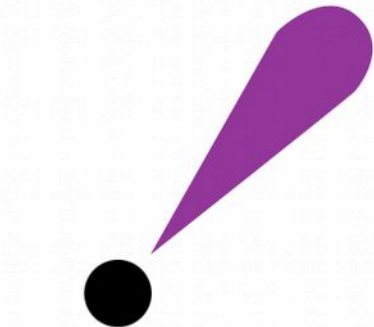
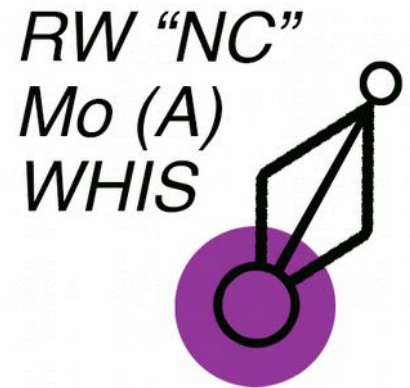
Oc R 4s



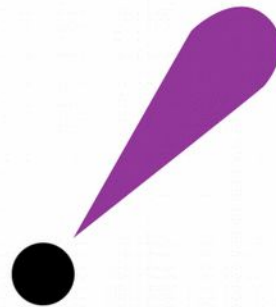
Oc (2) G 15s



Iso Y 6s



Oc (3) 15s



Iso R 4s



Al RW 6s



F G

Light Characteristics

Group 3

A

?

B

?

C

?

D

?

E

?

F

?

G

?

H

?

Light Characteristics

Group 3

A

Iso R 6s

B

Fl (2+1) G 6s

C

Fl (4) Y 15s

D

Fl (1+4+3) 45s

E

Oc (2) 15s

F

Fl R 4s

G

Fl (2) 12s

H

Oc G 4s

Light Characteristics

Group 4

A

?

B

?

C

?

D

?

E

?

F

?

G

?

H

?

Light Characteristics

Group 4

A

Fl R 2.5s

B

Q G

C

Fl Y 6s

D

Oc (3) 15s

E

Mo (A)

F

Fl R 4s

G

Al RW 10s

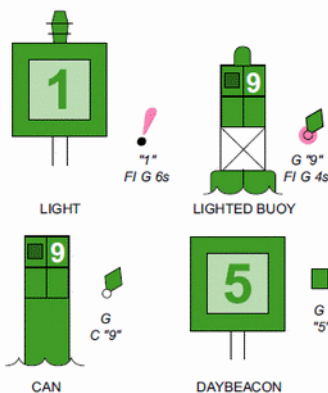
H

Fl (2+1) G 6s

LATERAL SYSTEM AS SEEN ENTERING FROM SEAWARD

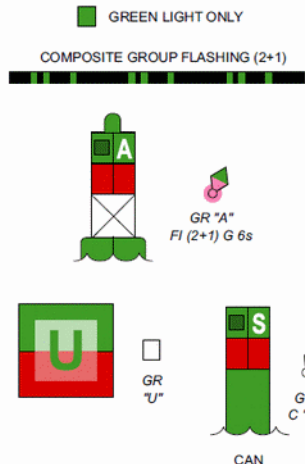
PORT SIDE ODD NUMBERED AIDS

- GREEN LIGHT ONLY
- FLASHING (2)
- FLASHING
- OCCULTING
- QUICK FLASHING
- ISO



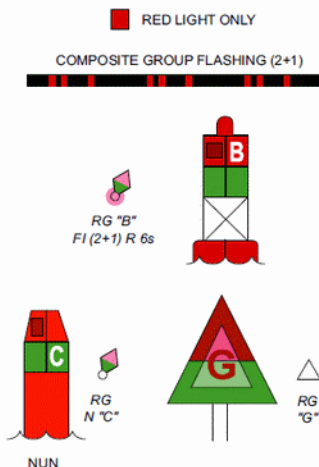
PREFERRED CHANNEL NO NUMBERS - MAY BE LETTERED

- PREFERRED CHANNEL TO STARBOARD
- TOPMOST BAND GREEN



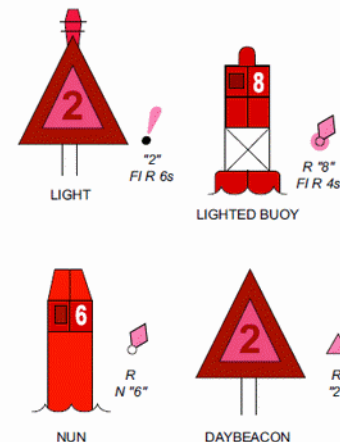
PREFERRED CHANNEL NO NUMBERS - MAY BE LETTERED

- PREFERRED CHANNEL TO PORT
- TOPMOST BAND RED



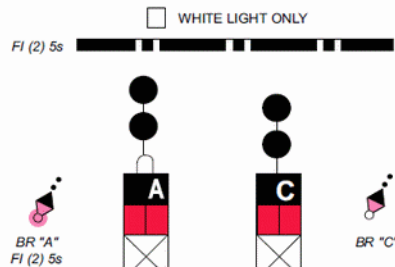
STARBOARD SIDE EVEN NUMBERED AIDS

- RED LIGHT ONLY
- FLASHING (2)
- FLASHING
- OCCULTING
- QUICK FLASHING
- ISO

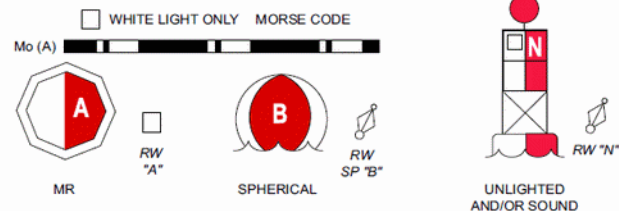


AIDS TO NAVIGATION HAVING NO LATERAL SIGNIFICANCE

ISOLATED DANGER NO NUMBERS - MAY BE LETTERED



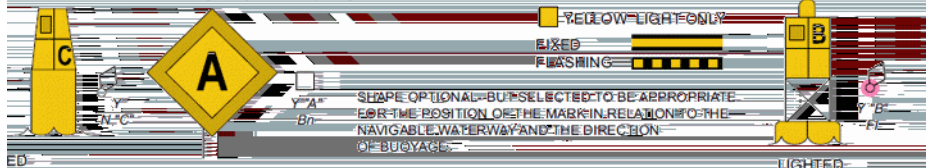
SAFE WATER NO NUMBERS - MAY BE LETTERED



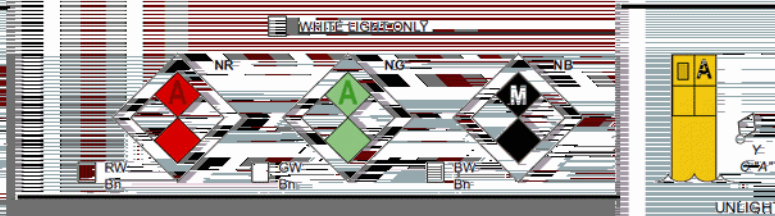
RANGE DAYBOARDS MAY BE LETTERED

KGW KWG KWB KBW KWR KRW KRB KBR KGB KBG KGR KRG

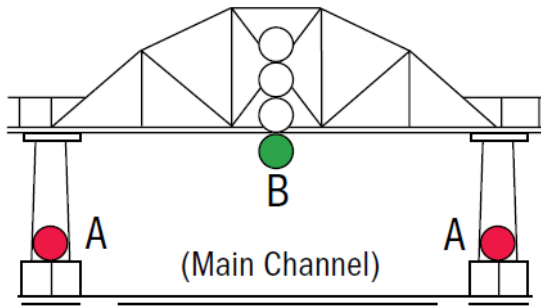
SPECIAL MARKS - MAY BE LETTERED



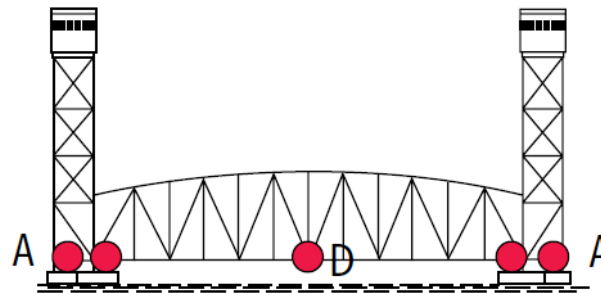
DAYBOARDS - MAY BE LETTERED



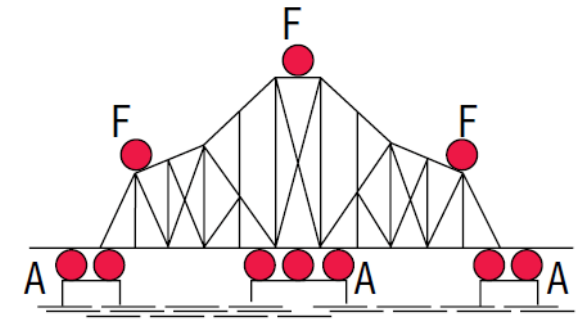
Bridges



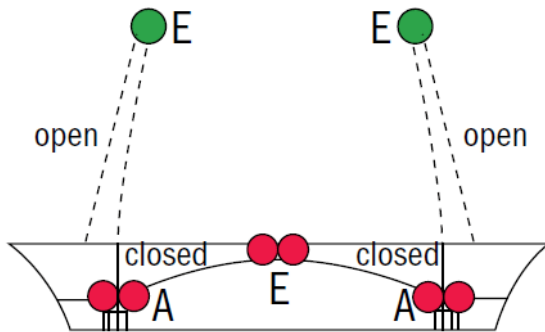
Fixed Bridge



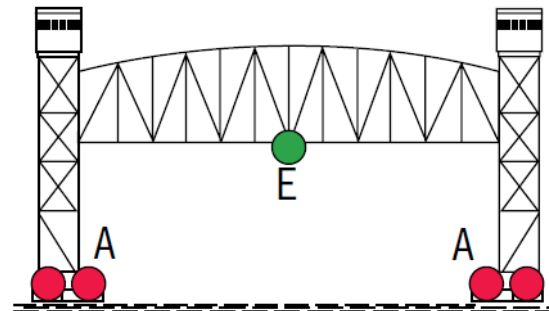
Vertical Lift Span Bridge
(closed)



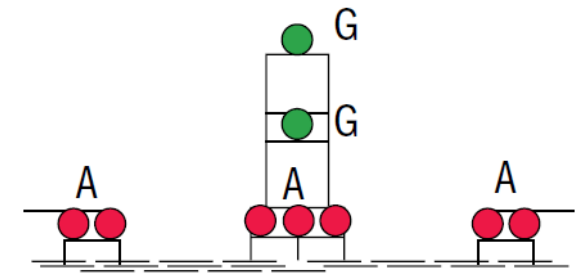
Double-Opening Swing Bridge
(closed)



Double Leaf (Lift) Bascule Bridge



Vertical Leaf Span Bridge
(open)



Double Opening Swing Bridge
(open)

Light List

Full details on all official ATONs can be found in the “Light List” (navcen.uscg.gov) including:

- Official number
- Official name
- Latitude & longitude
- Light pattern details (if lit)
- Height (if on land)
- Nominal Range (if lit)
- Physical description (e.g. “White Conical Tower with Red Stripe”, or “Steel Tripod with Mast”)
- Additional Comments

The Magnetic Compass



The Ship's Compass

The compass is usually mounted on a pedestal called a “binnacle”. It is directly in front of the helm, so the helmsman can steer by it.

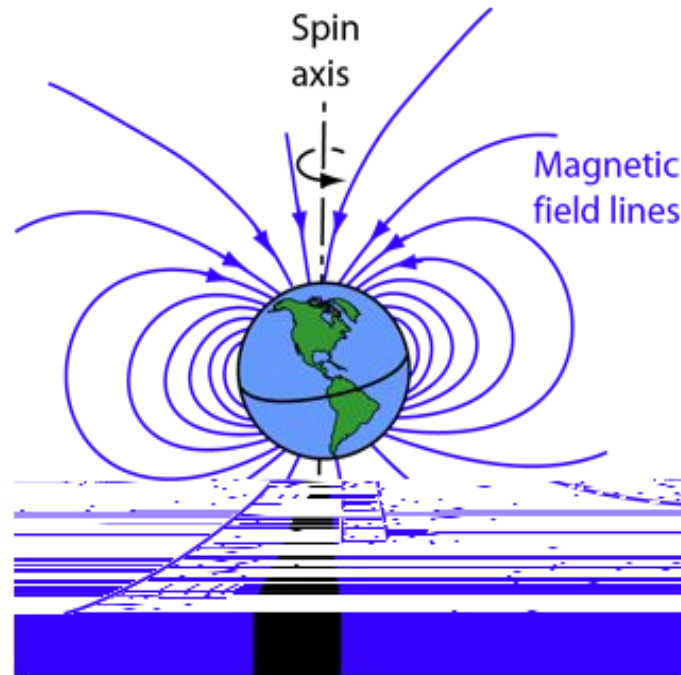
The compass contains a magnetized “card” floating in oil, weighted so it stays level, even if the ship is heeling. The heading is indicated by the numbers, written on the card, when they line up with the fixed “lubber's line”.



The Earth's Magnetic Field

The Earth's magnetic field is a three-dimensional vector field that changes in magnitude and direction over the surface of the Earth.

The magnetic field also varies slowly over time.



The Earth's Magnetic Field

The magnetic poles (created by the Earth's magnetic field) are not at the same locations as the geographic poles (defined by the rotation of the planet).

Note that compasses do not point at the magnetic poles, they point parallel to the Earth's magnetic field lines.

The Earth's Magnetic Field

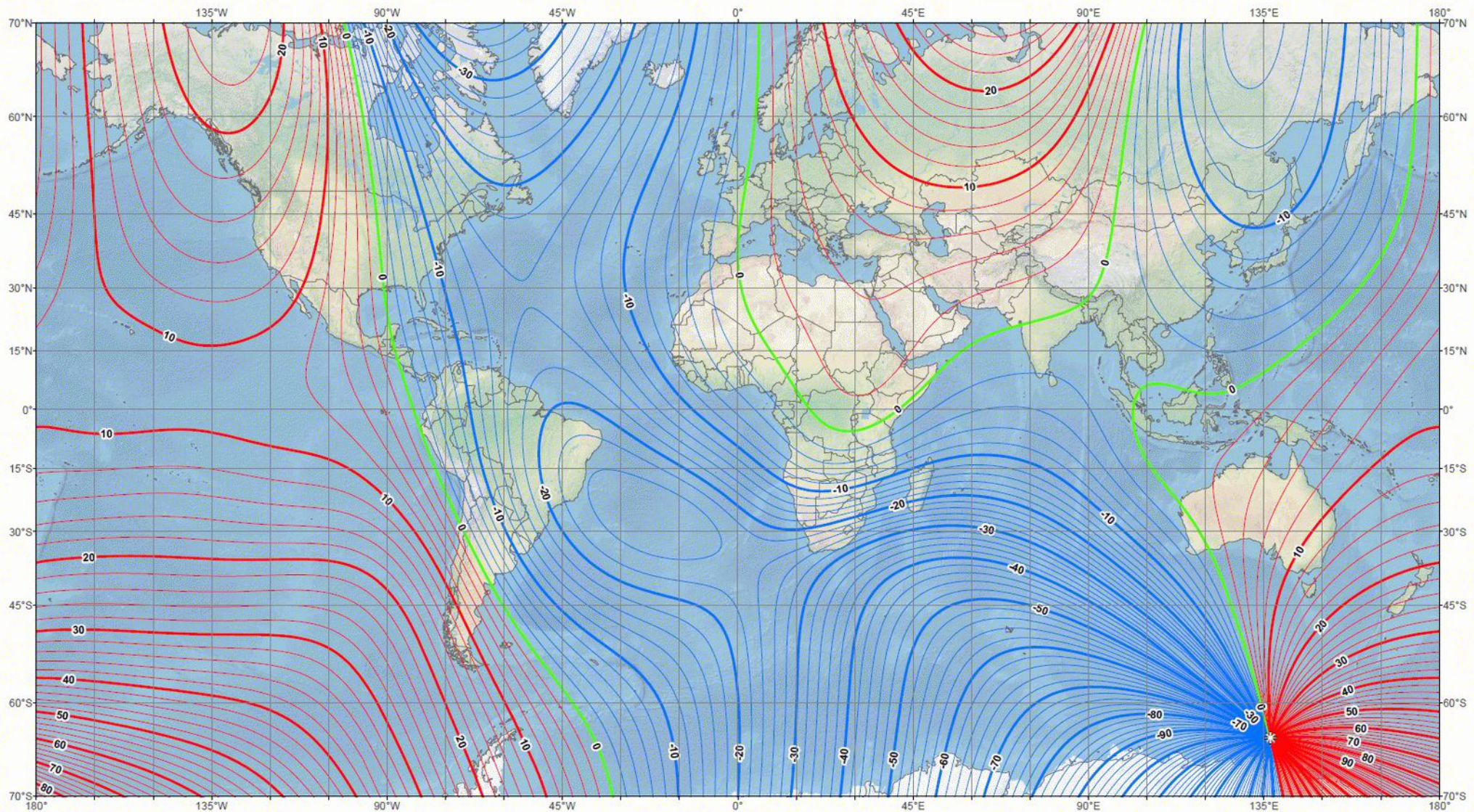
The geomagnetic field can be described at any given location on the surface of the Earth by two components.

- The vertical component (called the “dip”)
- The horizontal component (called the “declination” by land-lubbers, and called the “variation” by sailors).

Because of the dip, a compass needs to be kept level to give accurate readings.

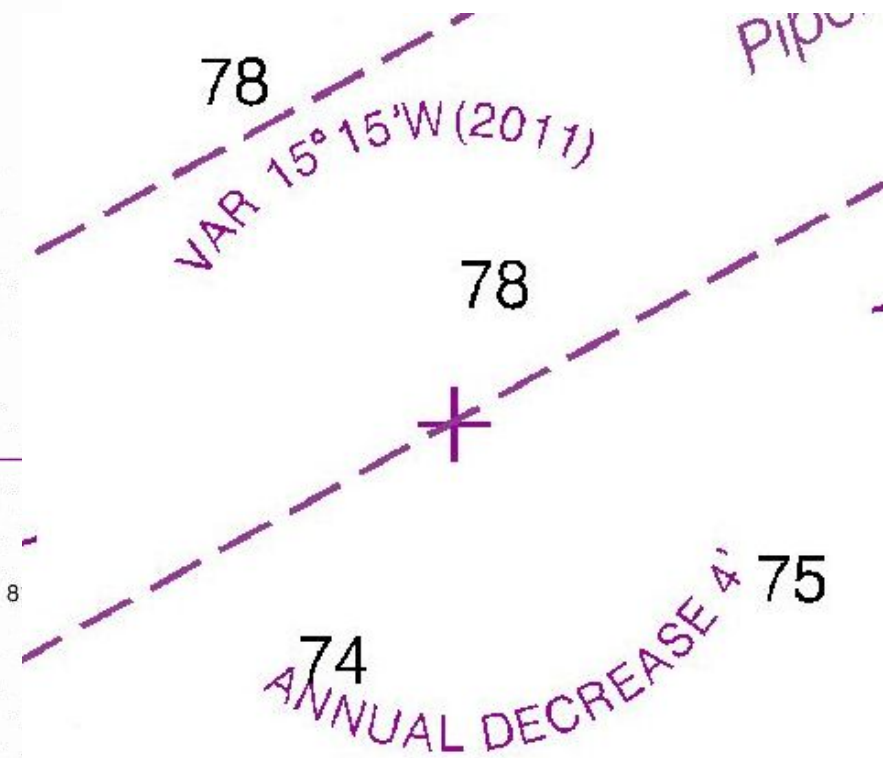
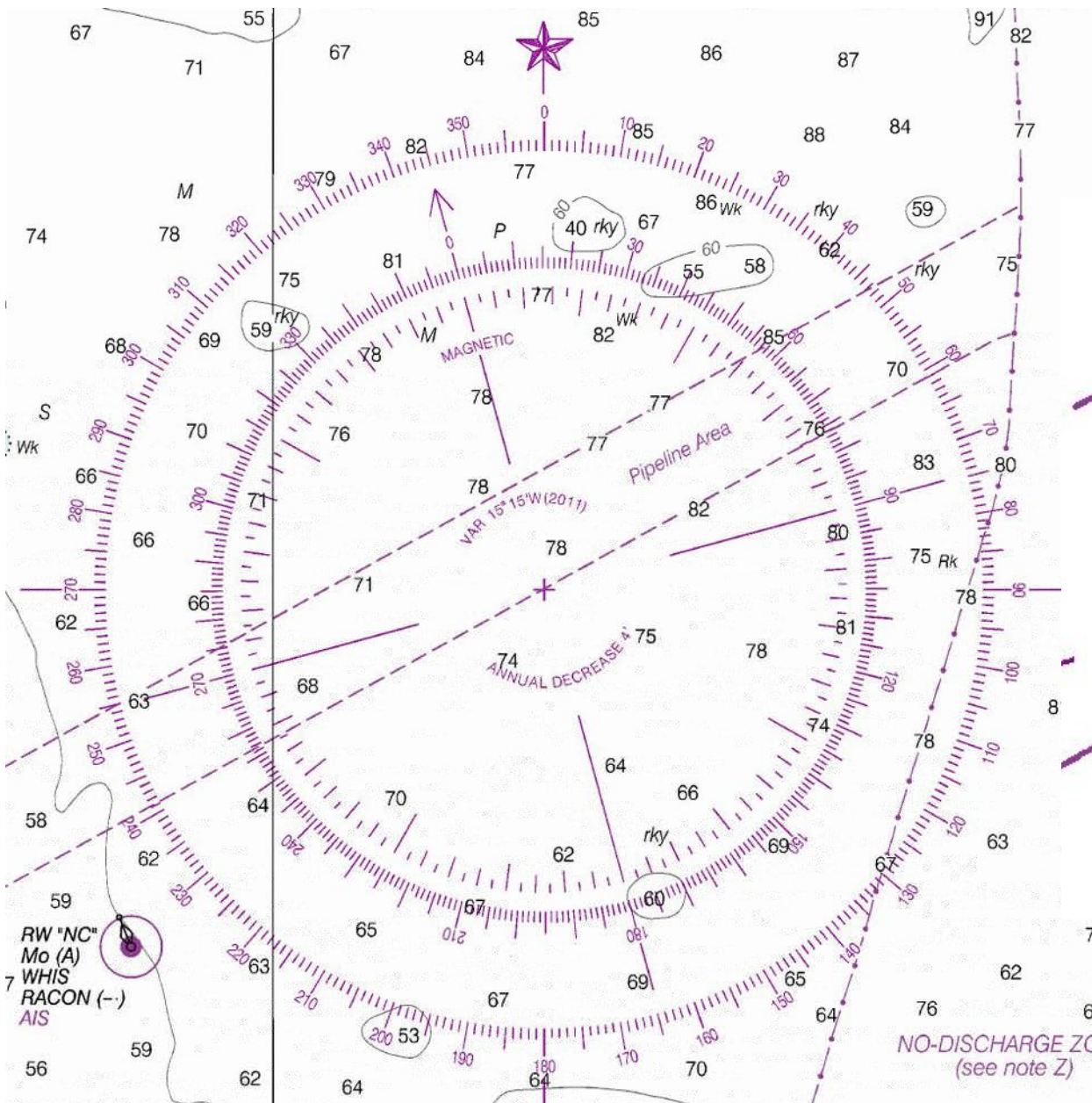
The Earth's Magnetic Field

US/UK World Magnetic Model - Epoch 2015.0
Main Field Declination (D)

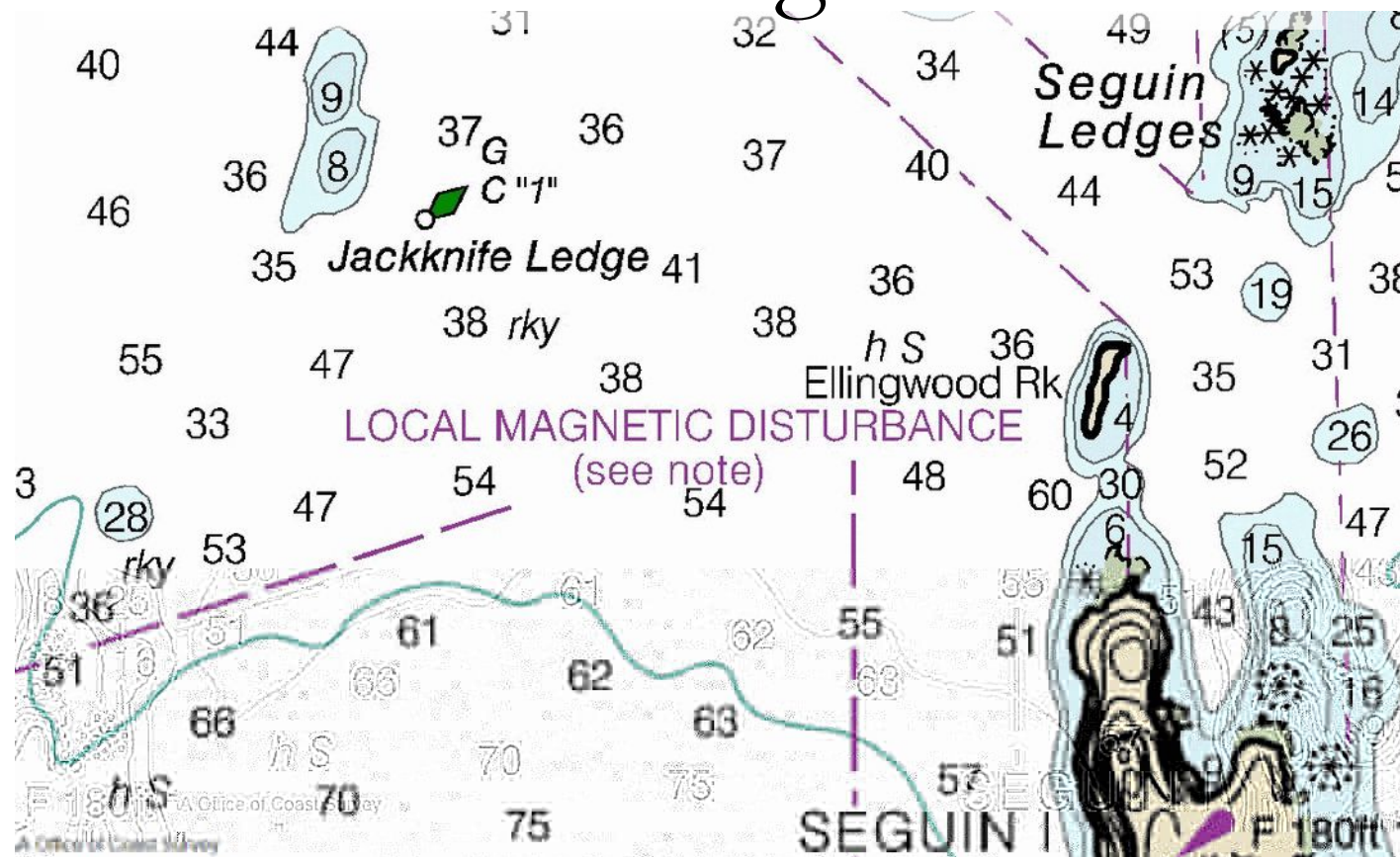


Compass Rose

Outside ring: True, Inside ring: Magnetic



The Earth's Magnetic Field



LOCAL MAGNETIC DISTURBANCE

Differences of as much as 8° from the normal variation have been observed in an area around Ellingwood Rock for approximately 1 nautical mile in all directions.

Magnetic Variation

Magnetic *variation* is the difference between a “true” direction (relative to the direction of the Geographic North Pole) and a “magnetic” direction (as indicated by a magnetic compass).

- Can be determined from compass rose on chart
- Can be calculated by a computer (GPS)
- Can be looked up in a variety of locations.

Bearings, courses and headings always need to be specified whether they are “true” or “magnetic”.

Magnetic Deviation

Magnetic *deviation* is the error in the compass reading caused by various local sources:

- Electric currents in nearby wires
- Nearby steel or other ferrous metals
- Nearby magnets (speakers, magnetized metal, etc)
- Poor compass manufacturing or installation

Deviation is a function of the boat's heading. Each compass on each boat may have a slightly different deviation function.

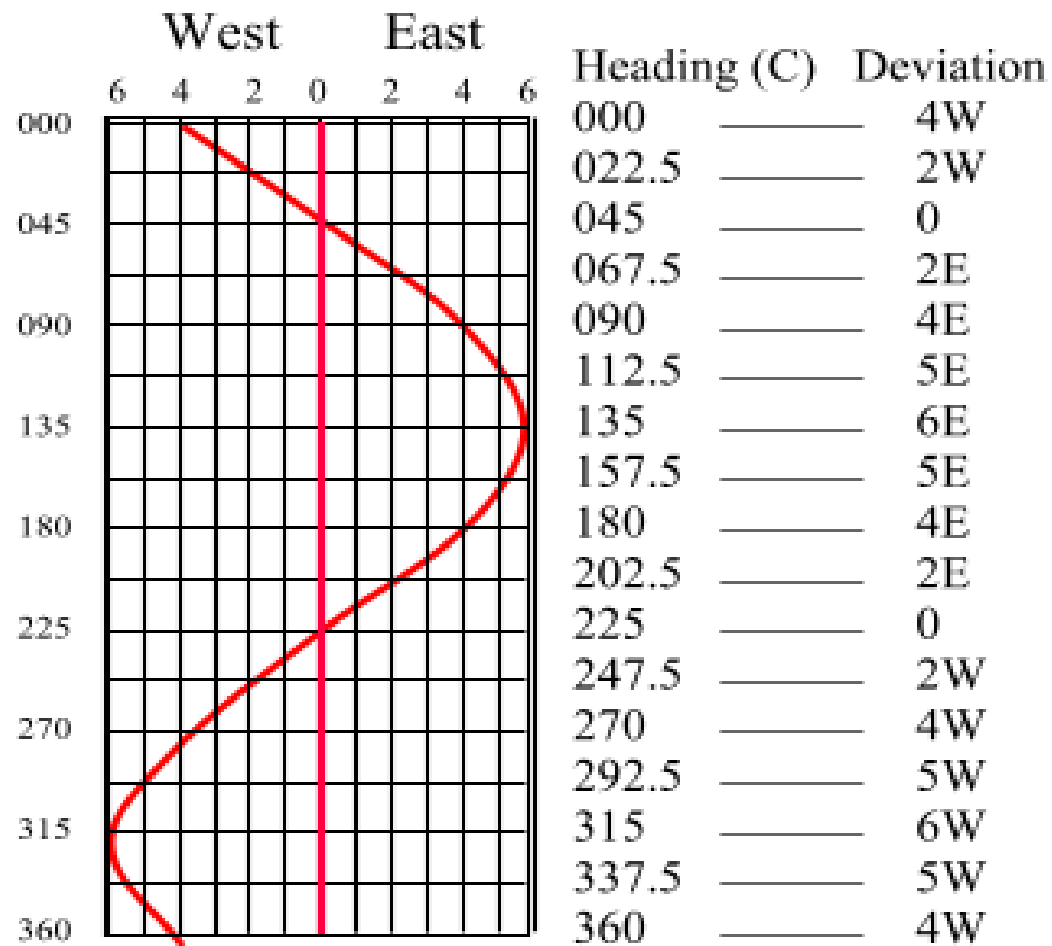
Magnetic Deviation

To determine deviation of a ship's compass:

- Point the boat in a known true heading (!)
- Take a compass reading
- Correct for magnetic variation
- Write down the difference
- Repeat for several different headings
- Plot the results on a graph
- Create a table of deviations for each heading

Magnetic Deviation

A Deviation Table



Compass Heading Correction

Can	Compass
Dead	Deviation
Men	Magnetic
Vote	Variation
Twice?	True

(at Elections)	(add East)
----------------	------------

Compass Heading Correction

Can	195°	Compass
Dead	3°E	Deviation
Men	198°	Magnetic
Vote	15°W	Variation
Twice?	183°	True

(at Elections)

(add East)

Compass Heading De-correction

True	True
Virtue	Variation
Makes	Magnetic
Dull	Deviation
Conversation	Compass
(add Whiskey)	(add West)

Compass Heading De-correction

True	46°	True
Virtue	15°W	Variation
Makes	61°	Magnetic
Dull	2°E	Deviation
Conversation	59°	Compass
(add Whiskey)		(add West)



Online Resources:

- navcen.uscg.gov
- charts.noaa.gov
- ngdc.noaa.gov/geomag/WMM/DoDWMM.shtml